



TRAILHEAD
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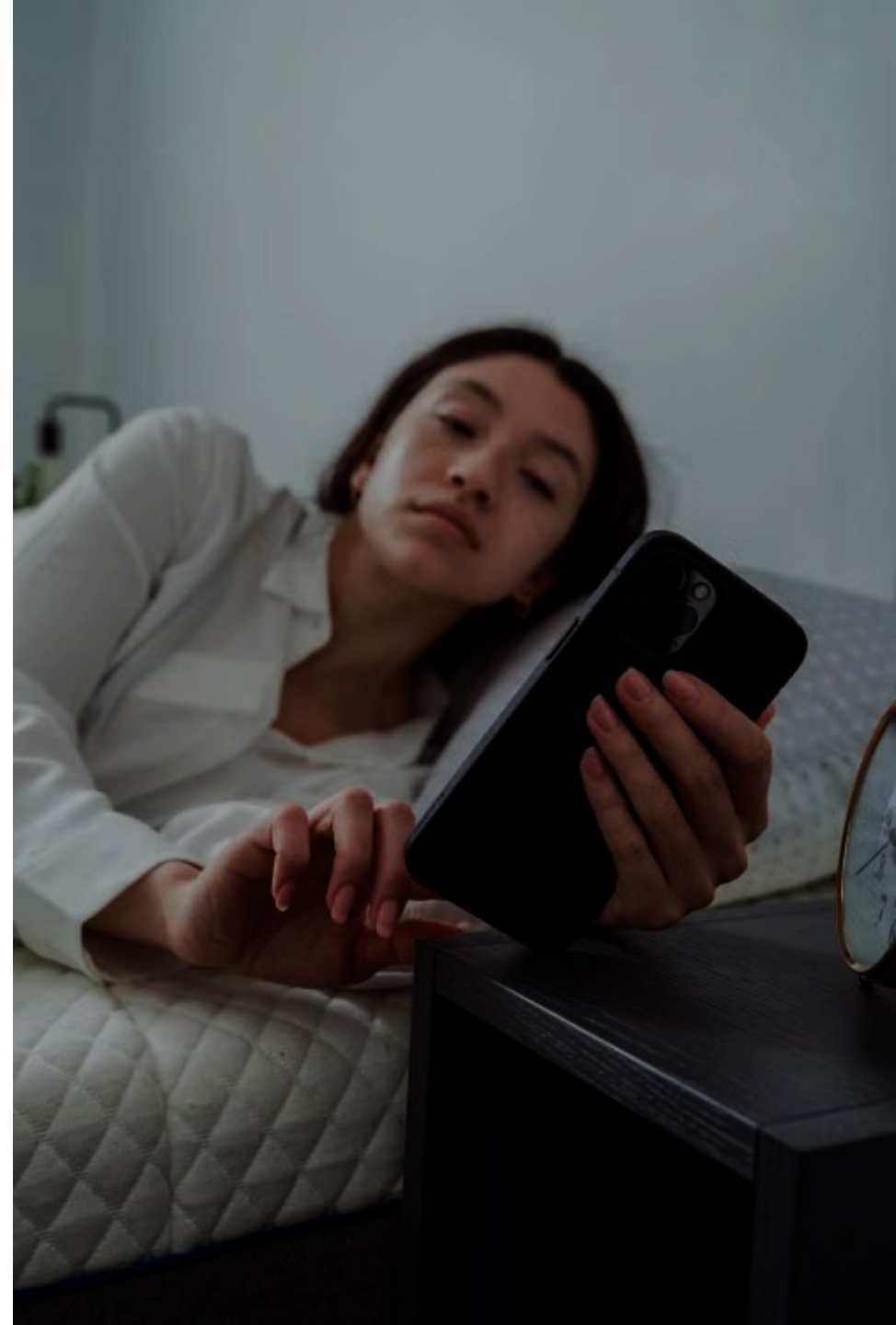
Pragmatic AI in .NET

10 AI Features Users Want in
Your .NET Apps



Jonathan "J." Tower

Saturday. 5:58 a.m.





Avalanche Lake Trail

40 trip
reports





She read
three.

"The
footbridge at
the creek is
gone. We
had to turn
around."

Report #14 of 40





Mile 2 is
where she
found out.

The answer was
there, but it was
buried in a
mountain of text.



Ten **shippable** AI features that solve problems your users **already have**, and the judgment to know **when to build each**.

What We'll Talk About



F01 Summarize



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What We'll Talk About



F01 Summarize



F02 Extract



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What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



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What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



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What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend

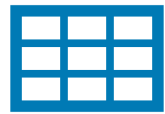


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What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend



F07 Route



What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend



F07 Route



F08 Detect



What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend



F07 Route



F08 Detect



F09 Approve



What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend



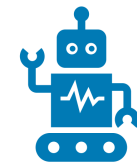
F07 Route



F08 Detect



F09 Approve



F10 Agent



What We'll Talk About



F01 Summarize



F02 Extract



F03 Sentiment



F04 Search



F05 RAG



F06
Recommend



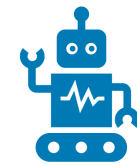
F07 Route



F08 Detect



F09 Approve



F10 Agent



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Jonathan "J." Tower

Founding Partner

🏆 13x Microsoft MVP in .NET & AI

⚙️ VP of .NET Foundation

🎙️ Blue Blazes podcast host



The Wrong Question and the Better One

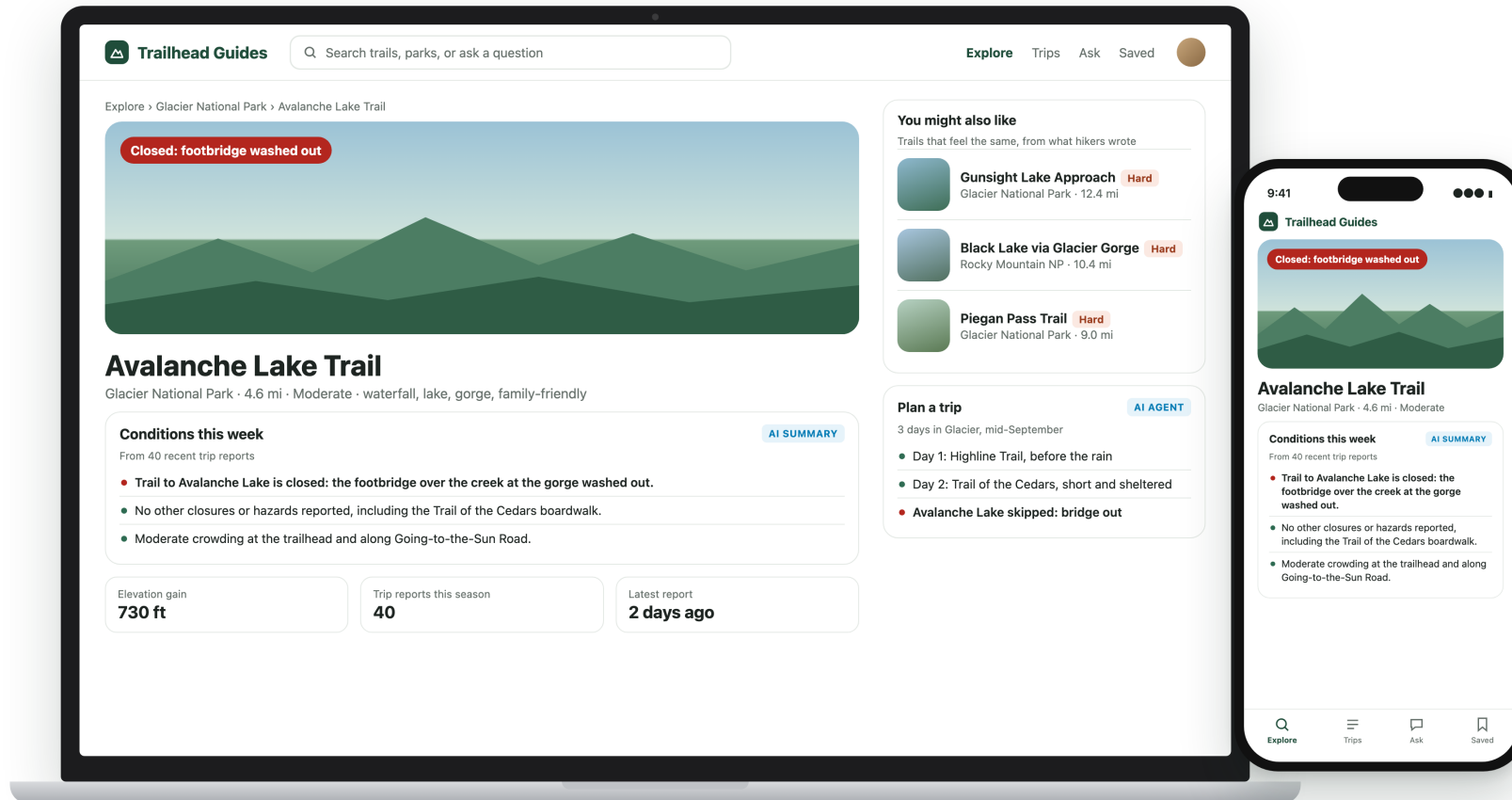
✗ "Where can we use AI?"

The Wrong Question and the Better One

✗ "Where can we use AI?"

✓ "What problems can AI *best* solve for our users?"

Meet Trailhead Guides



Local Models, Cloud When Needed

Local Models (~5 GB, no GPU):



llama3.2



phi3



nomic-embed-text

Cloud Model (my subscription):



Cloud models where it helps: F03, F05, F10

Every Model Is an HTTP Call

`localhost:11434 → /api/chat · /api/embed`

Every Model Is an HTTP Call

`localhost:11434 → /api/chat · /api/embed`

`...openai.azure.com/.../chat/completions + api-key`

Every Model Is an HTTP Call

`localhost:11434 → /api/chat · /api/embed`

`...openai.azure.com/.../chat/completions + api-key`

Same request shape · a URL and a key apart

One Interface, Local or Cloud

```
// Ollama on this laptop: free, private, small.
public static IChatClient Local(string model = "llama3.2") =>
    new OllamaApiClient(new Uri("http://localhost:11434"), model);

// Azure OpenAI on Foundry: same interface, different constructor.
public static IChatClient Cloud(string deployment) =>
    new AzureOpenAIClient(new Uri(Endpoint), new ApiKeyCredential(Key))
        .GetChatClient(deployment)
        .AsIChatClient();

// Embeddings stay local in every demo.
public static IEmbeddingGenerator<string, Embedding<float>> Embedder() =>
    new OllamaApiClient(new Uri("http://localhost:11434"), "nomic-embed-text");
```

One Interface, Local or Cloud

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```

F01 • Summarization



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Turn a long document into the
short version its reader needs.



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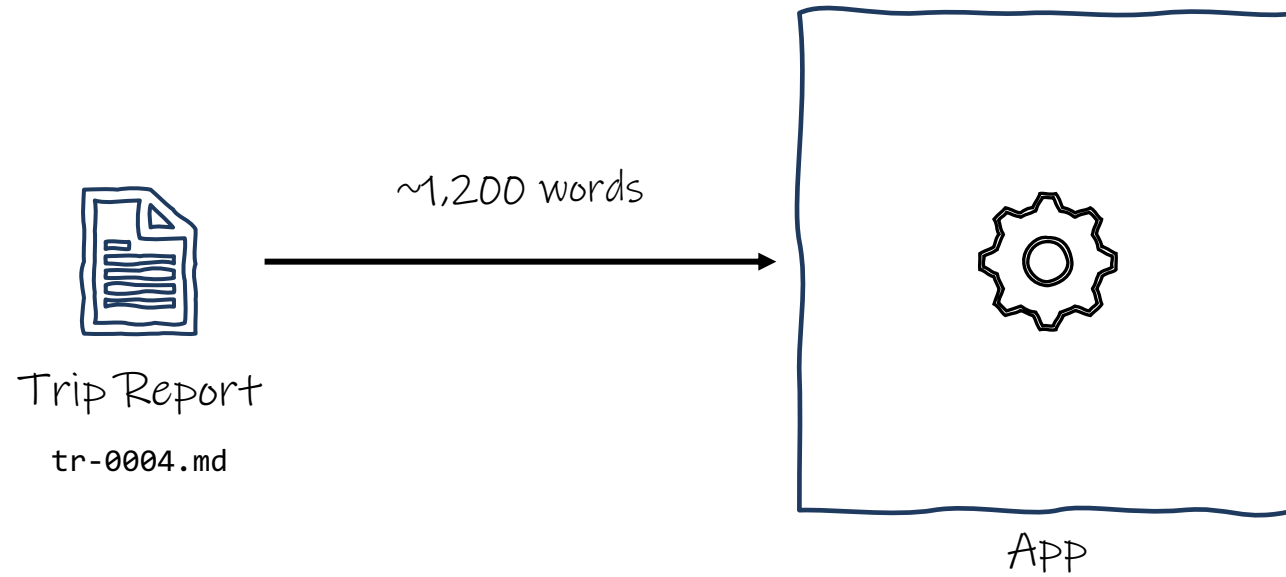
F01 · How It Works



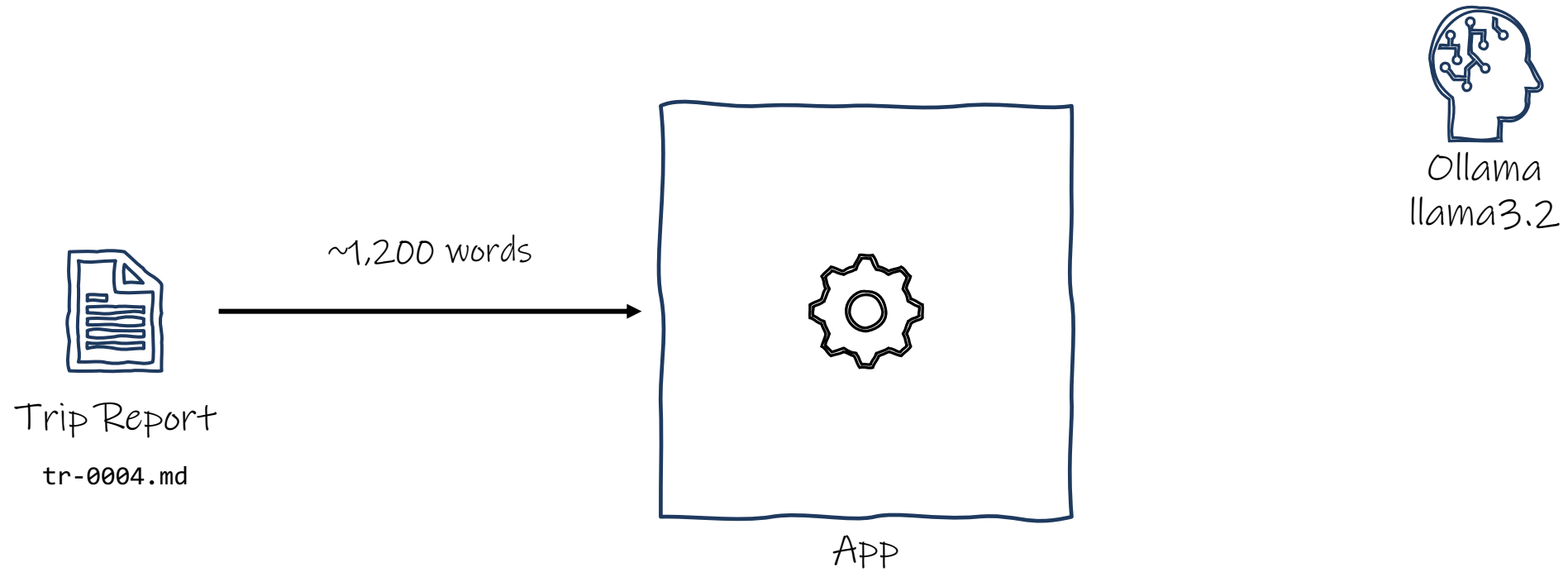
Trip Report

tr-0004.md

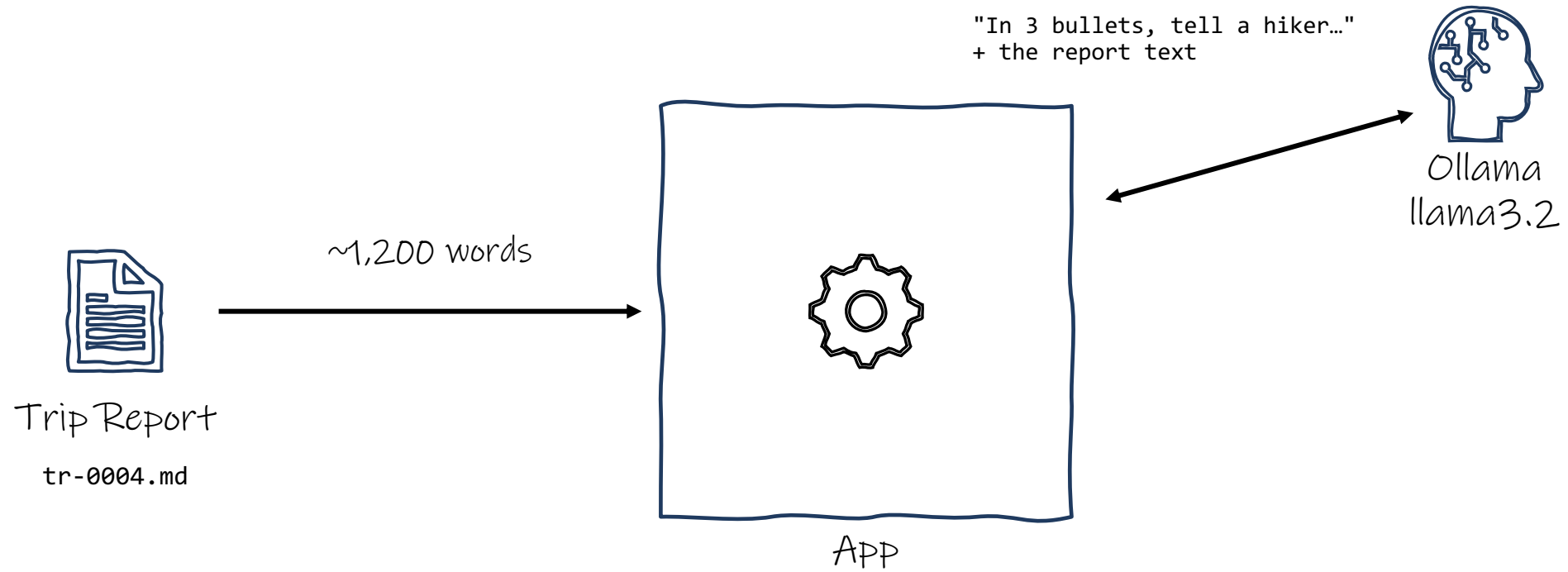
F01 · How It Works



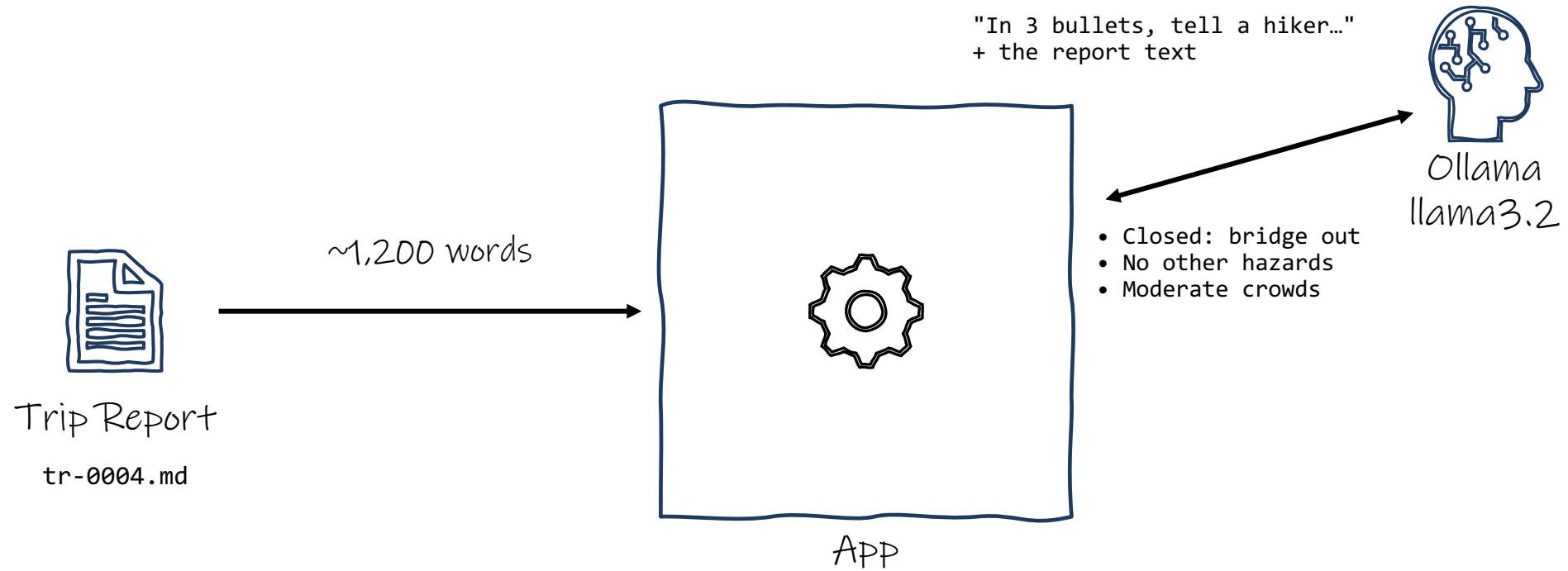
F01 · How It Works



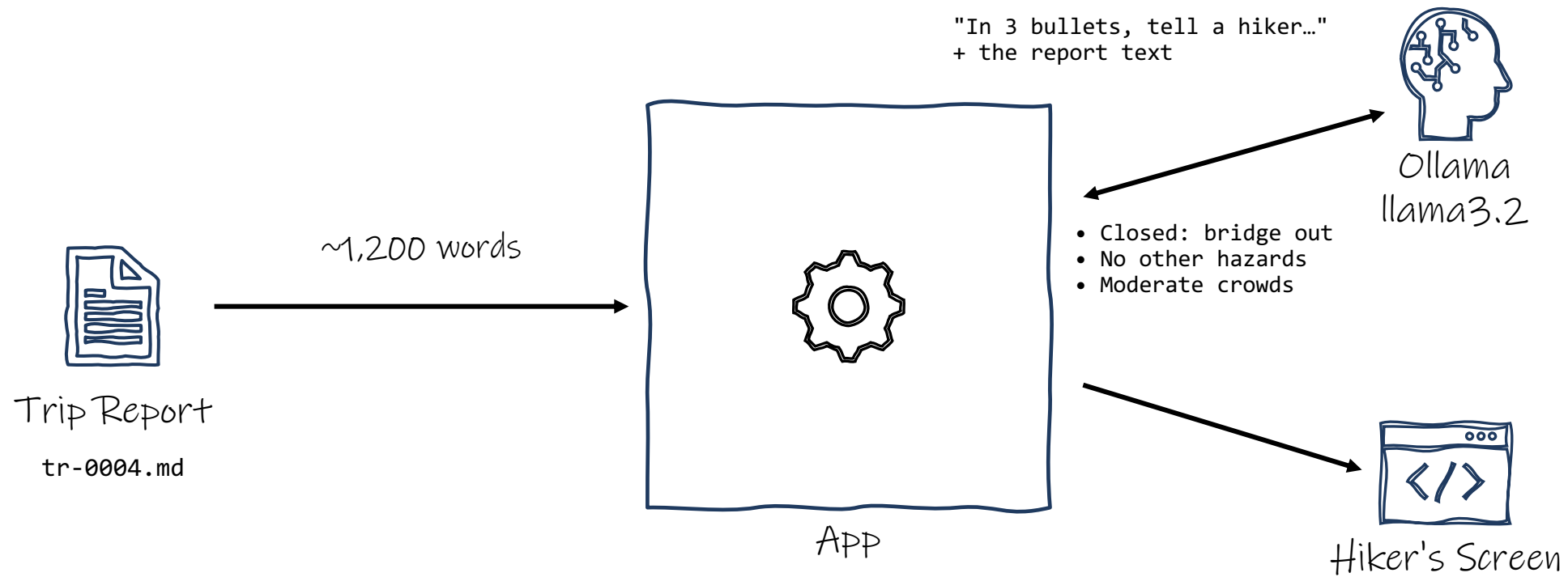
F01 · How It Works



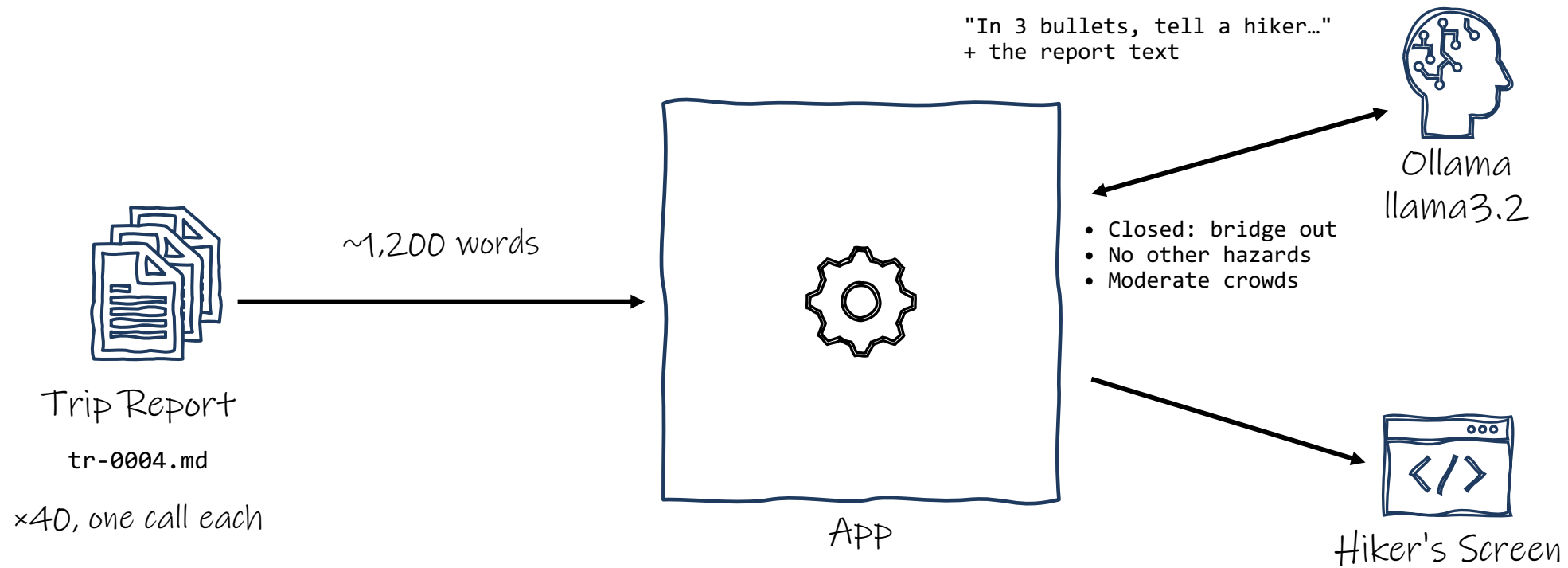
F01 · How It Works



F01 · How It Works



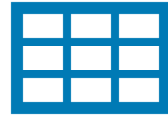
F01 · How It Works



F02 • Extraction



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Pull facts out of free-form text
into **fields your code can query**.



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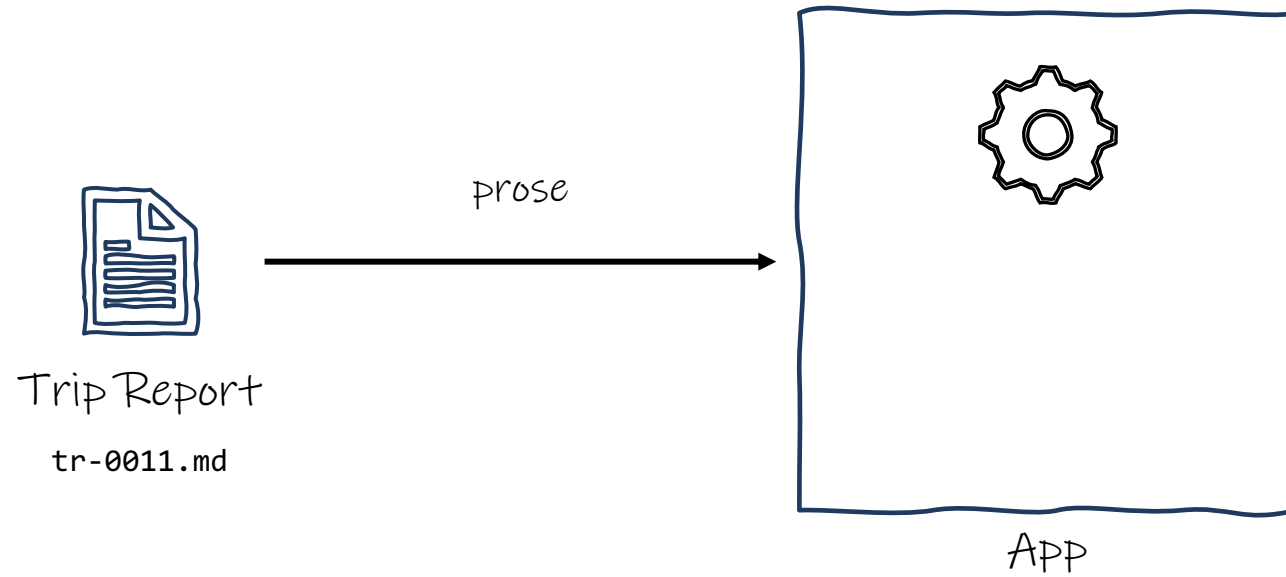
F02 · How It Works



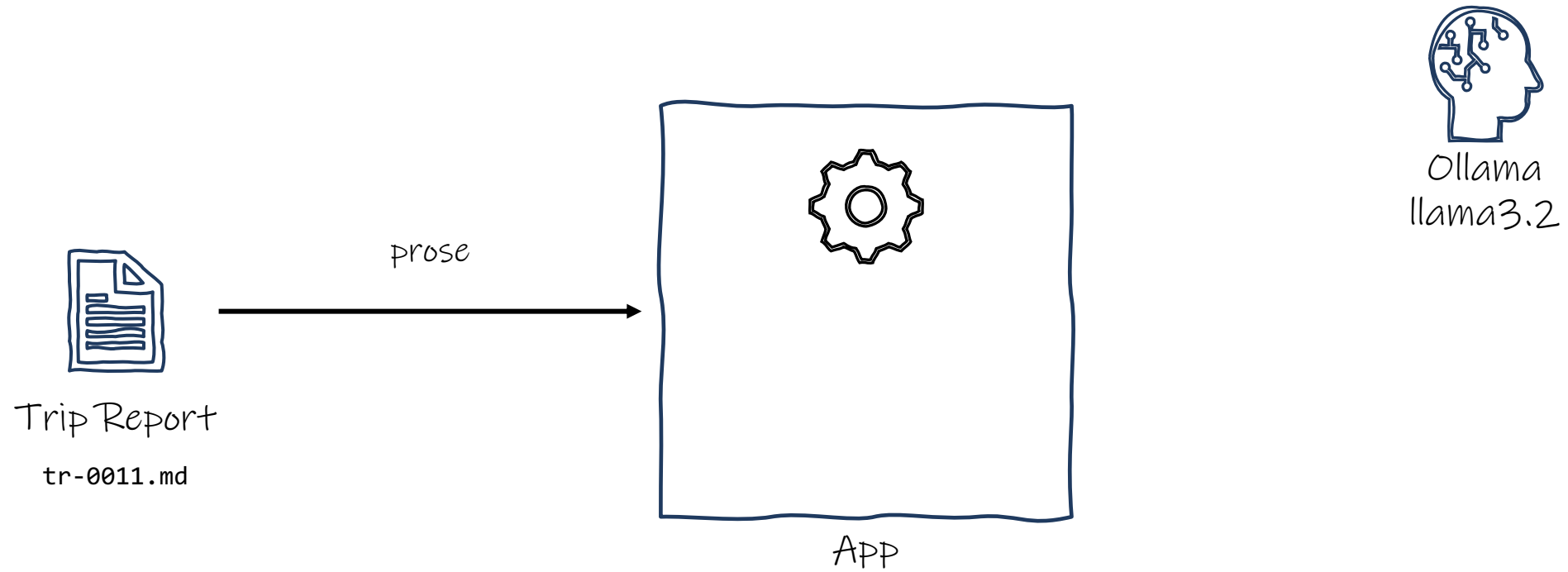
Trip Report

tr-0011.md

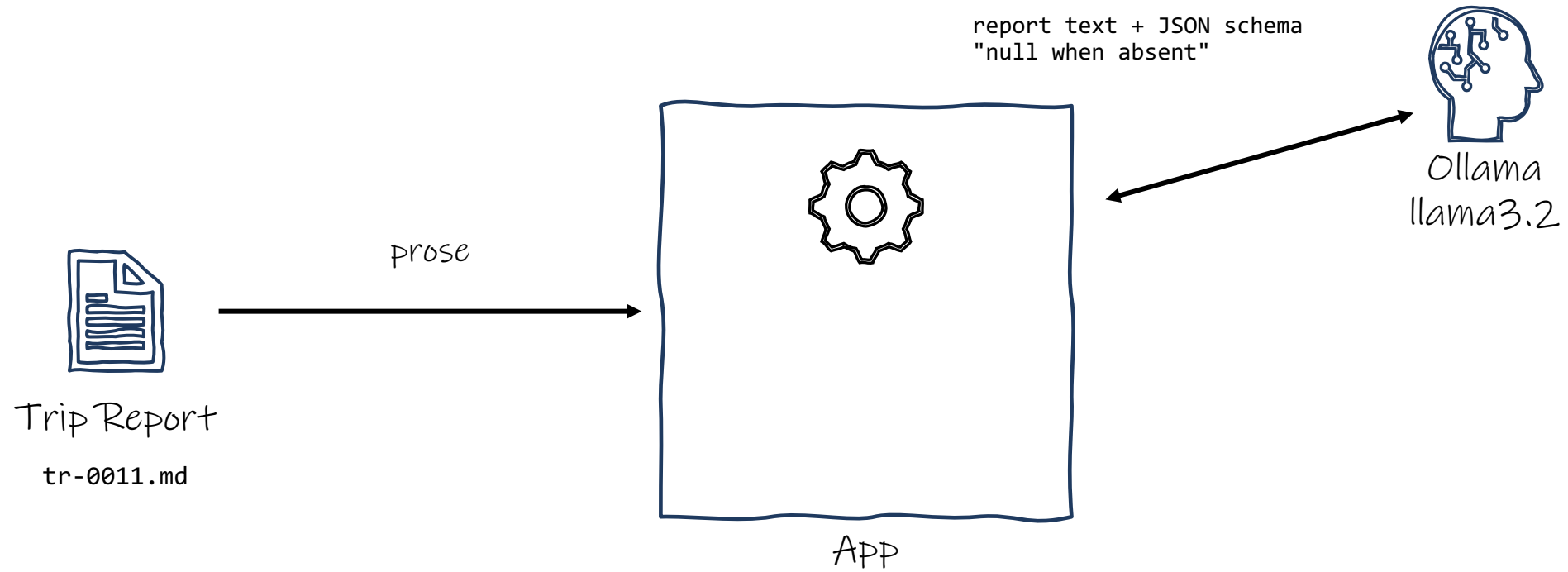
F02 · How It Works



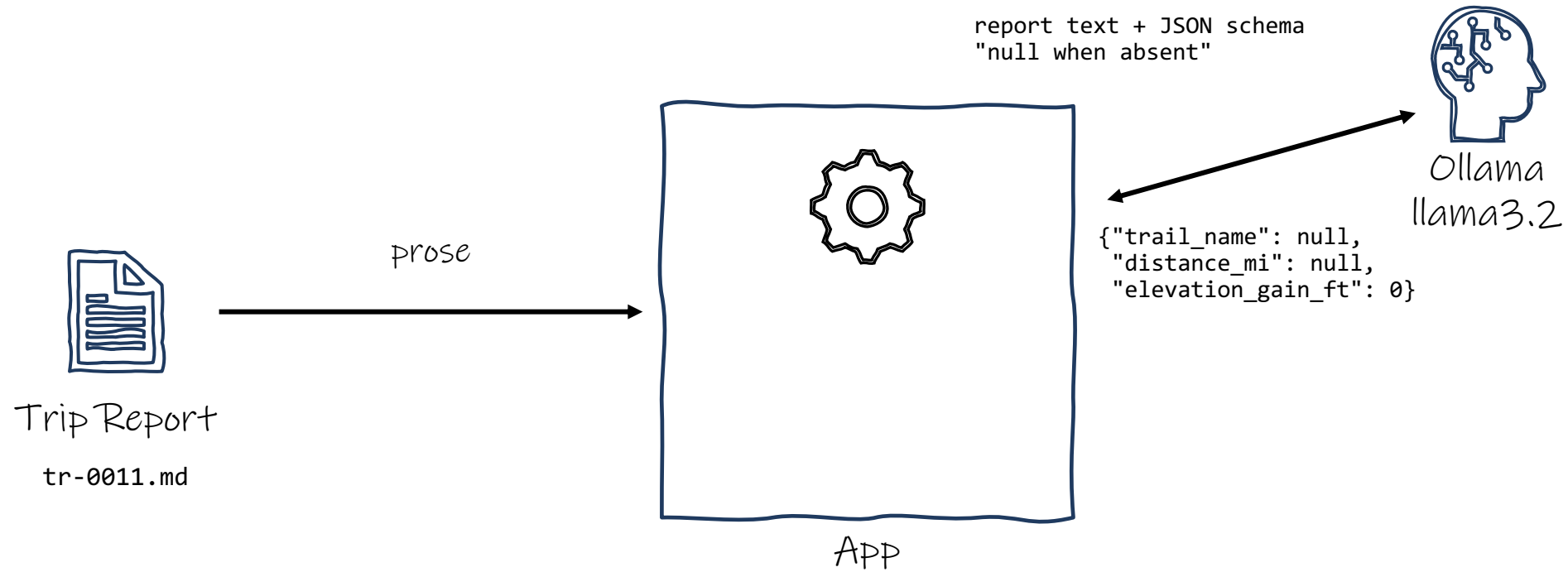
F02 · How It Works



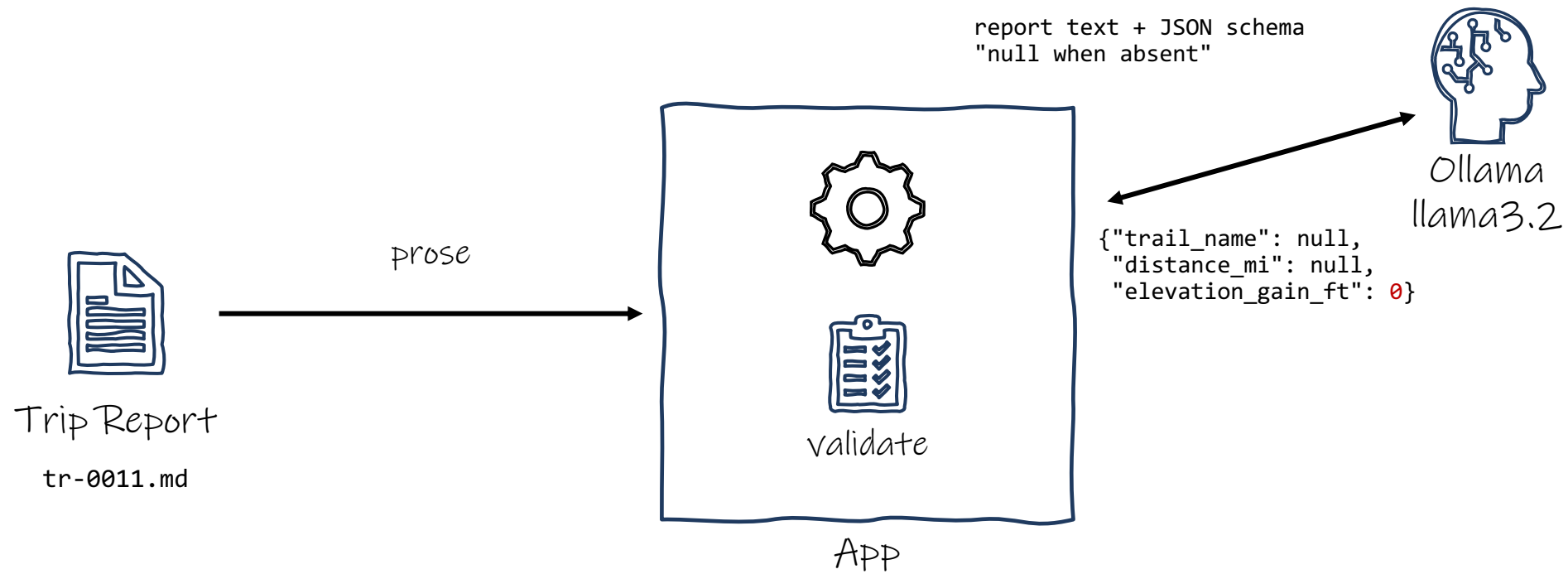
F02 · How It Works



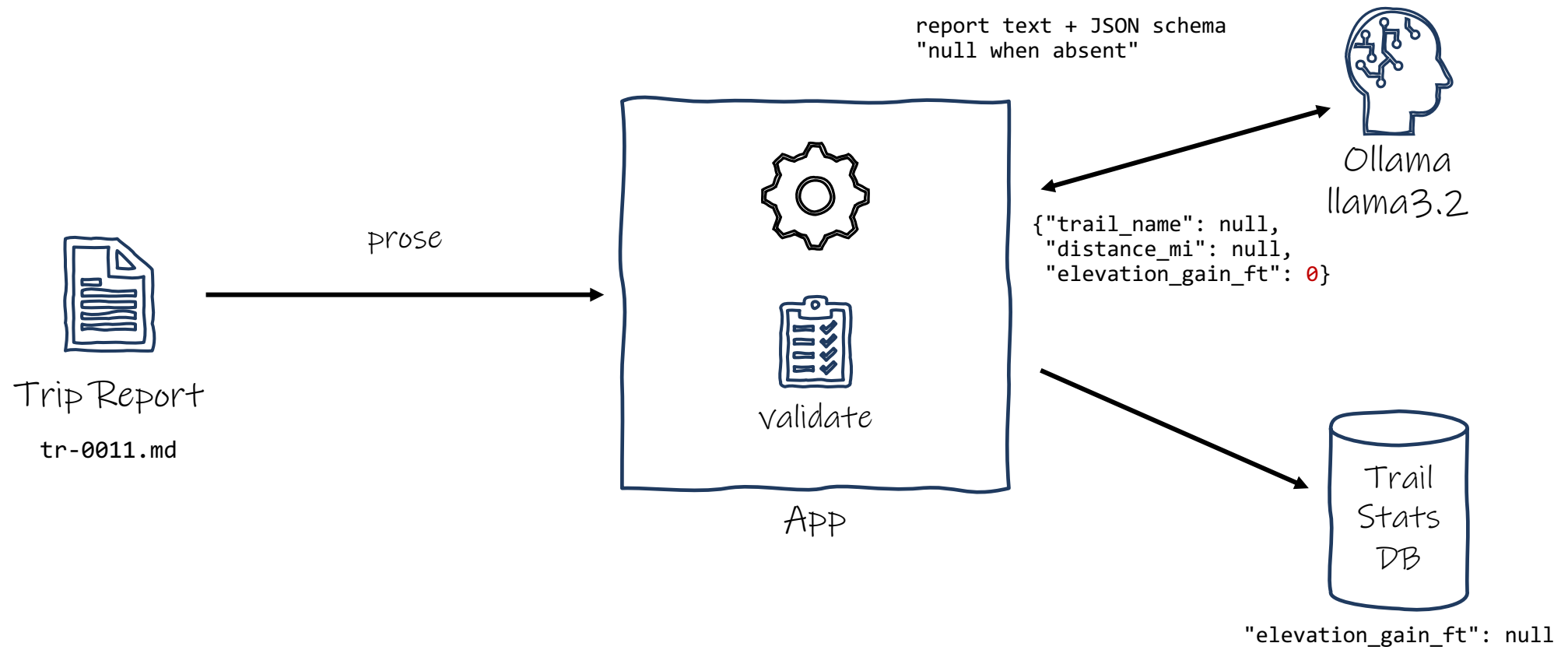
F02 · How It Works



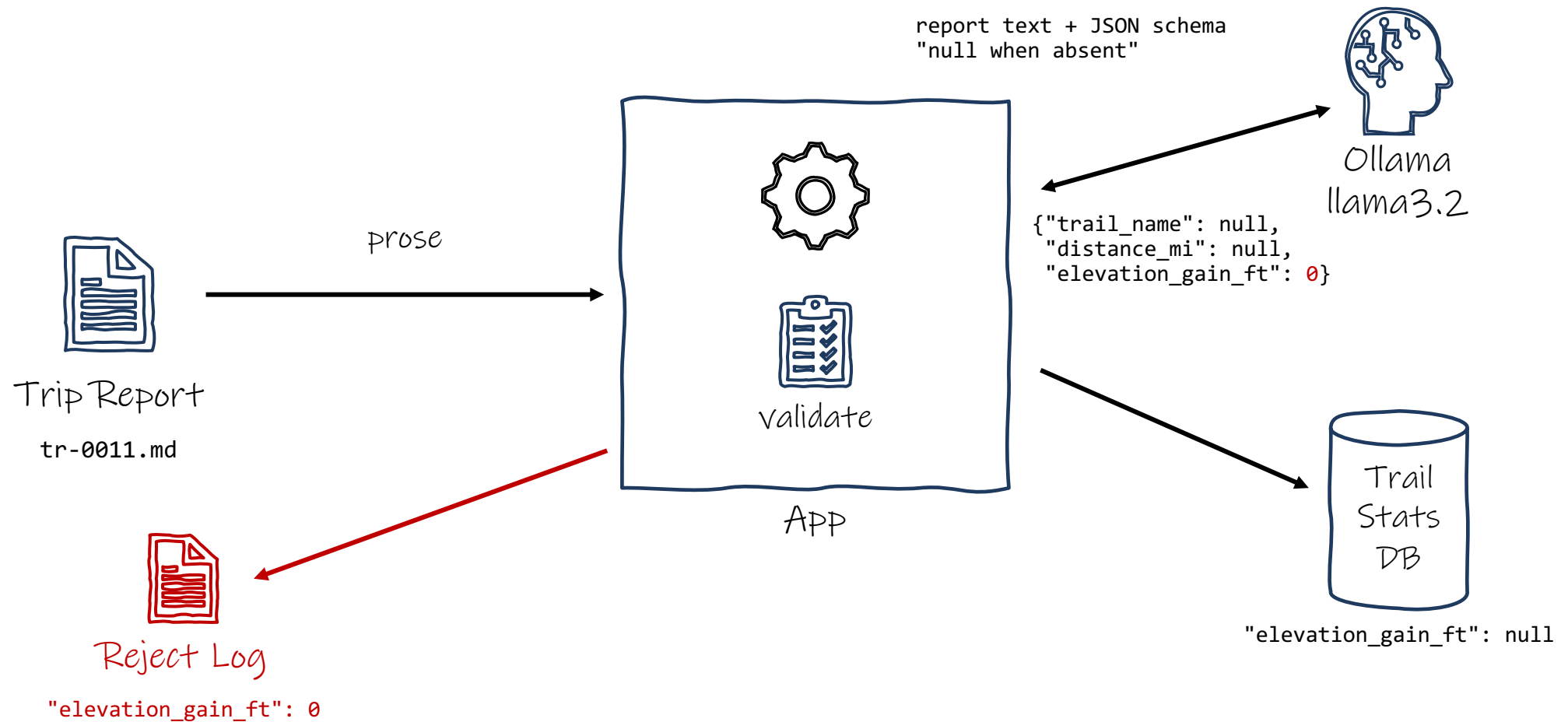
F02 · How It Works



F02 · How It Works



F02 · How It Works



F02 · The Call

```
var report = await Load(file);

// Prose in, typed object out. The record is the schema.
var response = await client.GetResponseAsync<TripFacts>($$$$
    Extract the trail facts from this trip report.
    Use null for any field the report does not state,
    and empty arrays when nothing applies. Do not guess.

    {report}
    $$$);
TripFacts raw = response.Result;

// The schema guarantees it parses, not that it is true.
var verdicts = Validate(raw, report);
Show("what we would store", Clean(raw, verdicts));
```

F02 · The Call

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var report = await Load(file);

// Prose in, typed object out. The record is the schema.
var response = await client.GetResponseAsync<TripFacts>($"""
    Extract the trail facts from this trip report.
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TripFacts raw = response.Result;

// The schema guarantees it parses, not that it is true.
var verdicts = Validate(raw, report);
Show("what we would store", Clean(raw, verdicts));
```

F02 · The Schema

```
record TripFacts(  
    string? TrailName,  
    string? Park,  
    string? DateHiked,  
    [property: Description("Round-trip distance in miles, as stated " +  
        "in the report. null, never 0, if the report gives no " +  
        "distance. Never estimate.")]  
    double? DistanceMi,  
    double? ElevationGainFt,  
    string[] Wildlife,  
    string[] Conditions,  
    string[] Hazards);
```

F02 · The Schema

```
record TripFacts(  
    string? TrailName,  
    string? Park,  
    string? DateHiked,  
    [property: Description("Round-trip distance in miles, as stated " +  
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    double? DistanceMi,  
    double? ElevationGainFt,  
    string[] Wildlife,  
    string[] Conditions,  
    string[] Hazards);
```

F02 · The Validator

```
static List<Verdict> Validate(TripFacts f, string sourceText) =>
[
    // A name the source text never contains does not get to be a fact.
    Grounded("trail_name", f.TrailName, sourceText),
    Grounded("park", f.Park, sourceText),
    ValidDate("date_hiked", f.DateHiked),

    // 0 is the dangerous near-miss: it is a value, so a pipeline stores it
    // and nothing downstream ever questions it. The honest answer is null.
    InRange("distance_mi", f.DistanceMi, max: 100, unit: "mi"),
    InRange("elevation_gain_ft", f.ElevationGainFt, max: 20000, unit: "ft"),
];
```

F02 · The Validator

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    // A name the source text never contains does not get to be a fact.
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];
```

DEMO · F02 Extraction

F03 · Sentiment



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Read text and judge **the feeling behind it**: positive, negative, or mixed.



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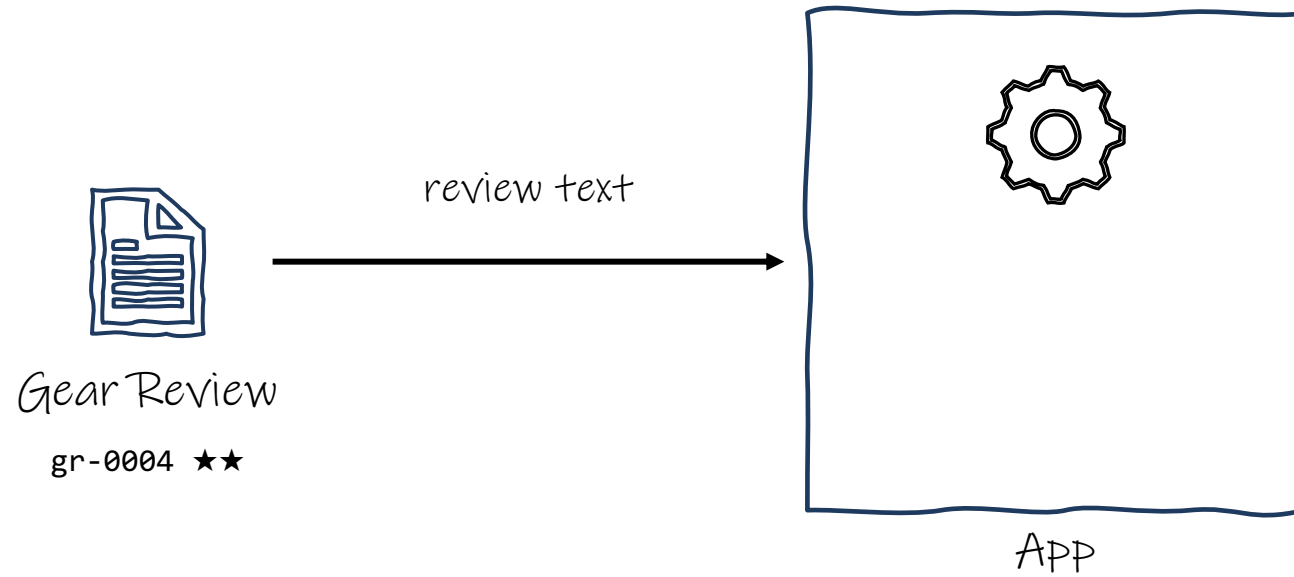
F03 · How It Works



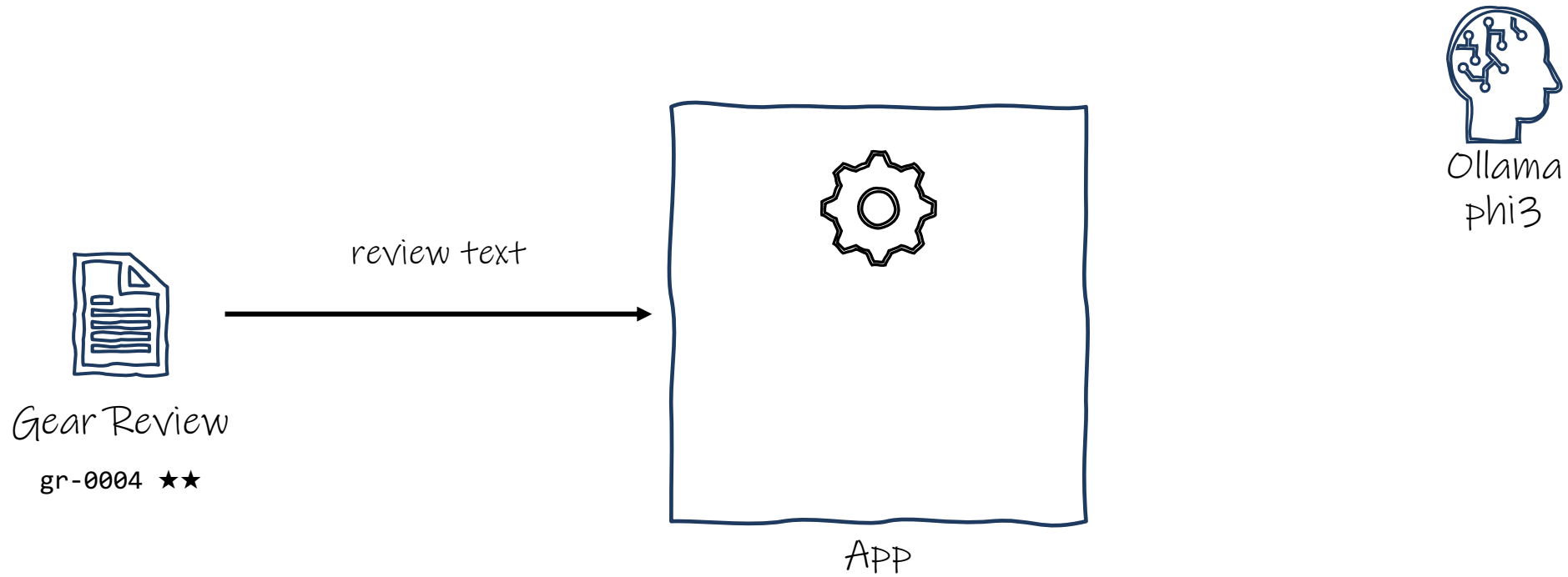
Gear Review

gr-0004 ★★

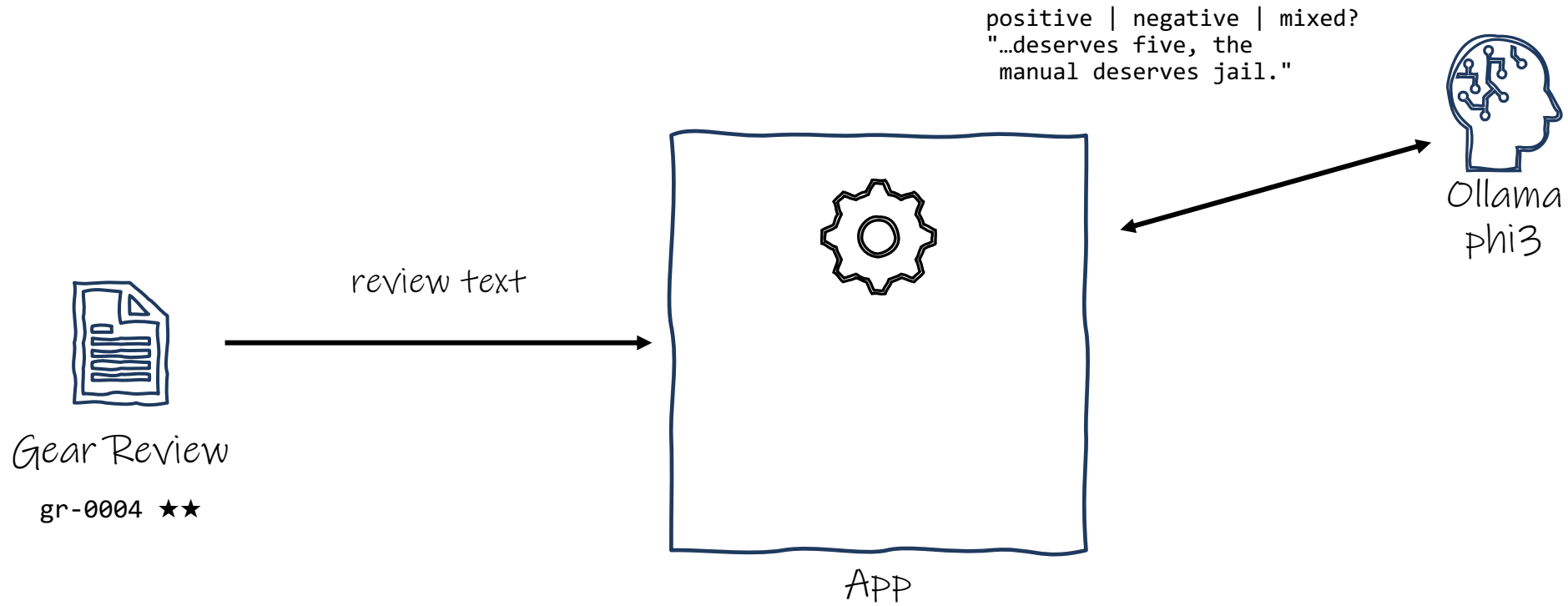
F03 · How It Works



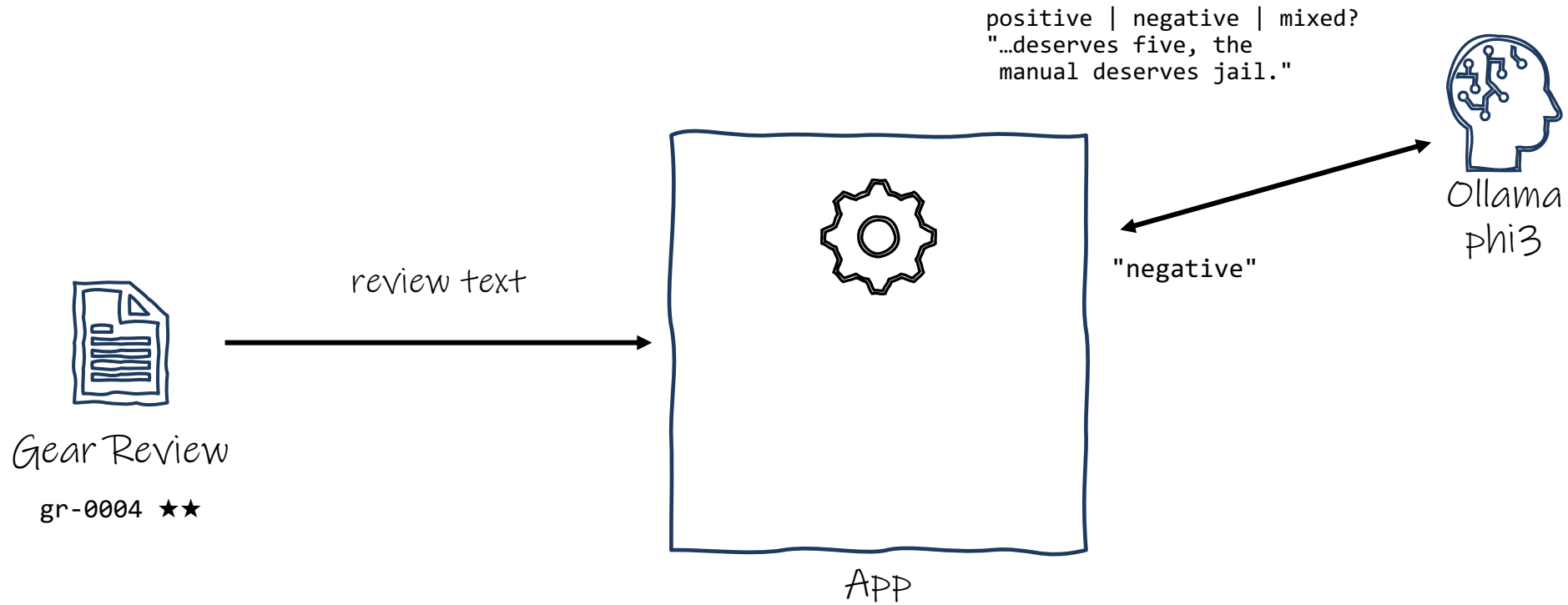
F03 · How It Works



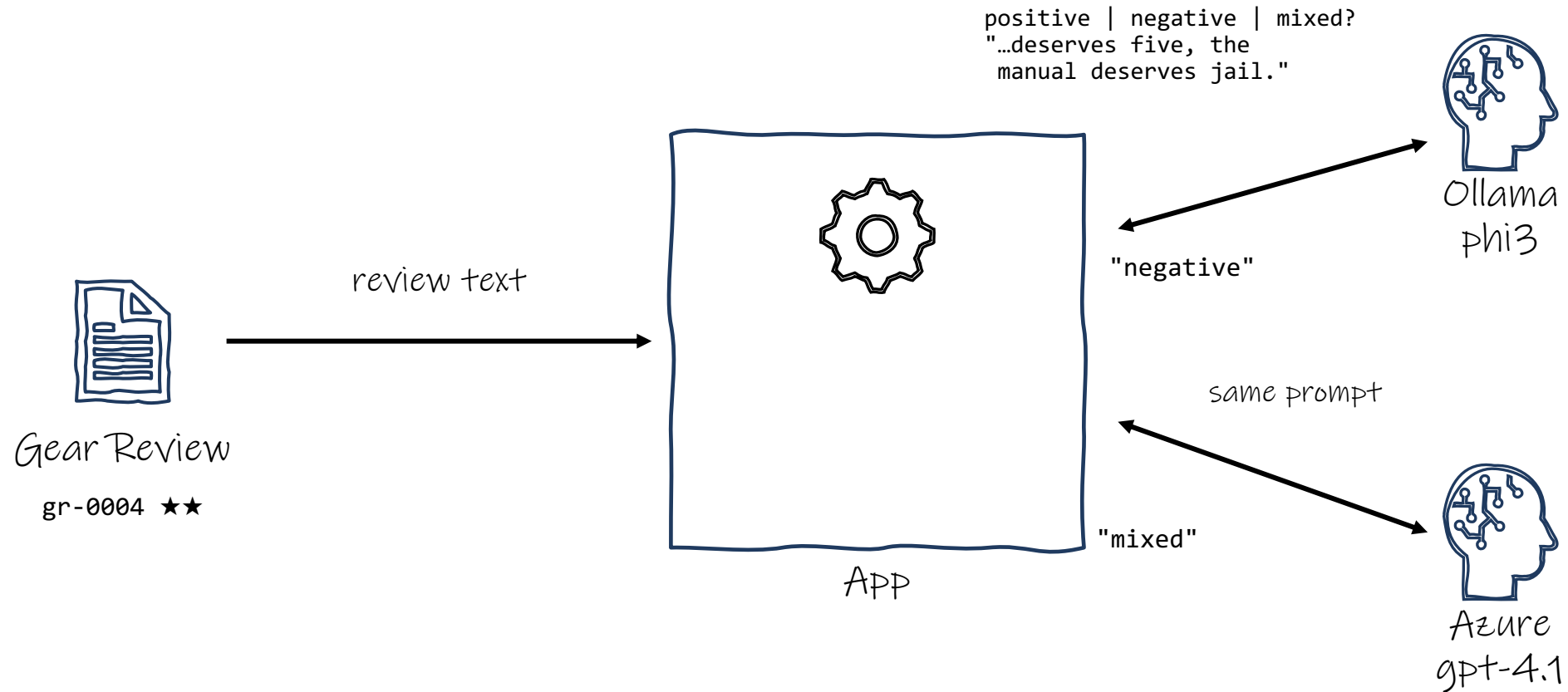
F03 · How It Works



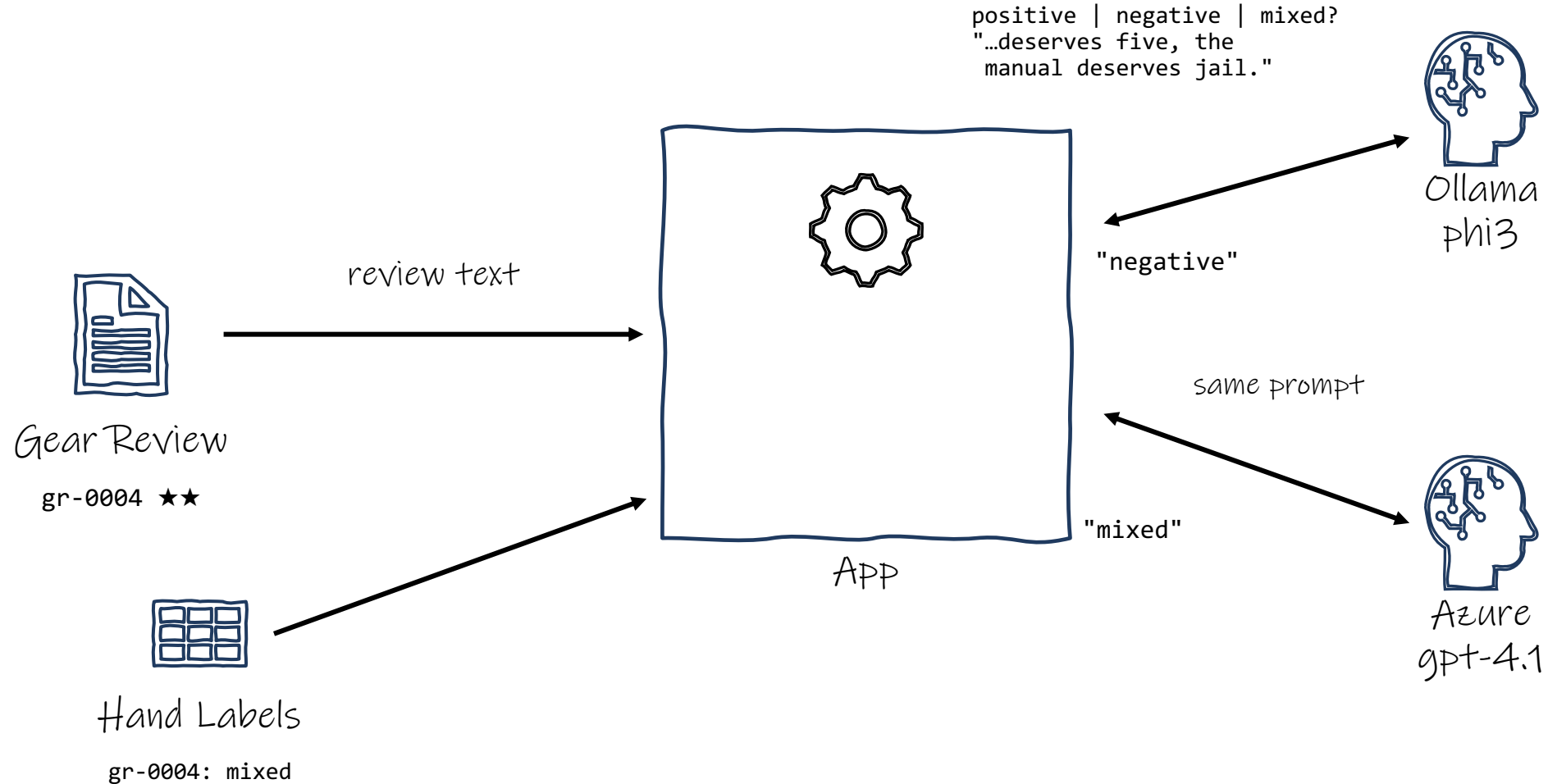
F03 · How It Works



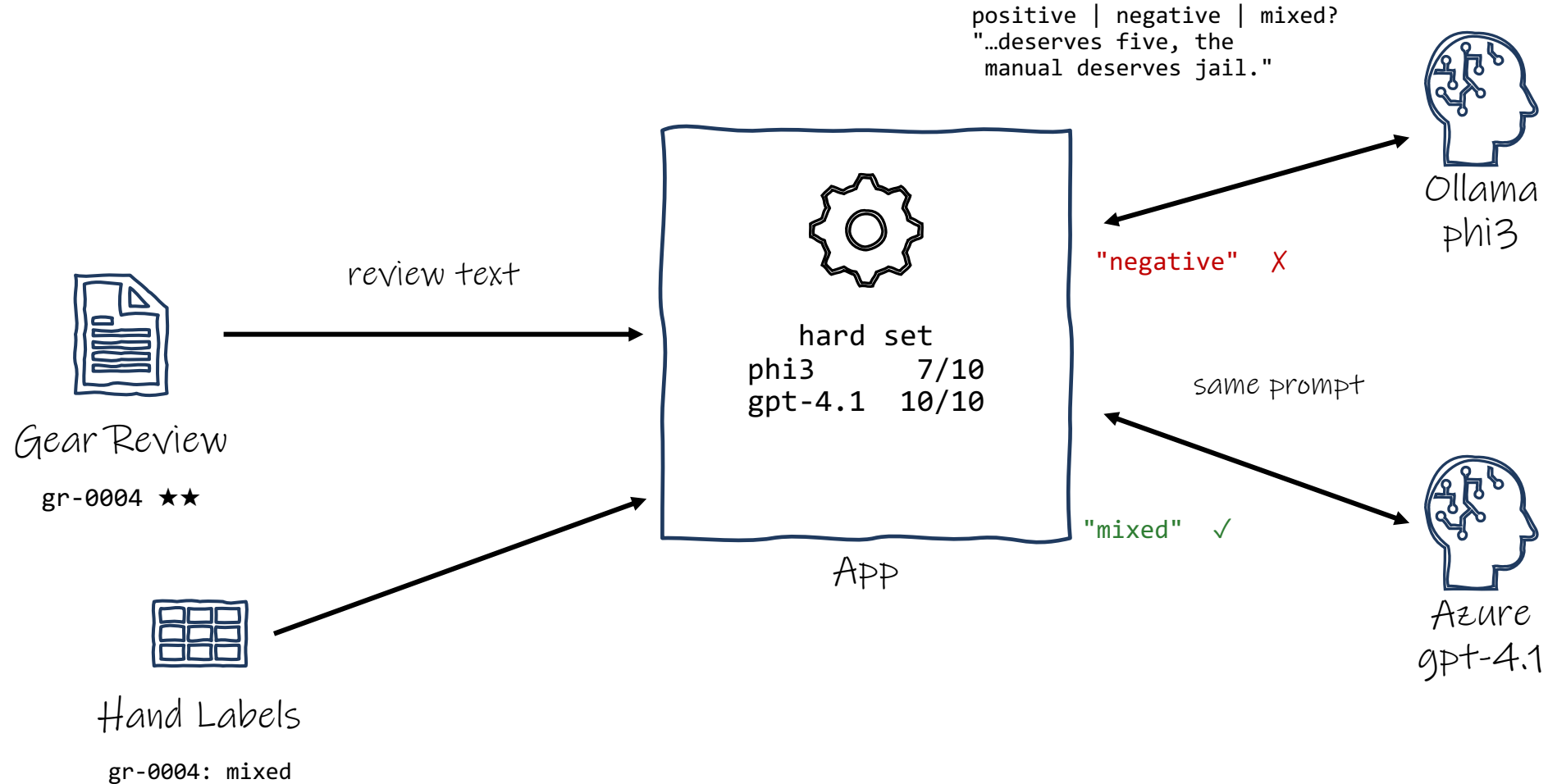
F03 · How It Works



F03 · How It Works



F03 · How It Works



F04 · Semantic Search



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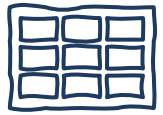
Search by what words **mean**,
not by which words appear.



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F04 · How It Works

1 · Index, once

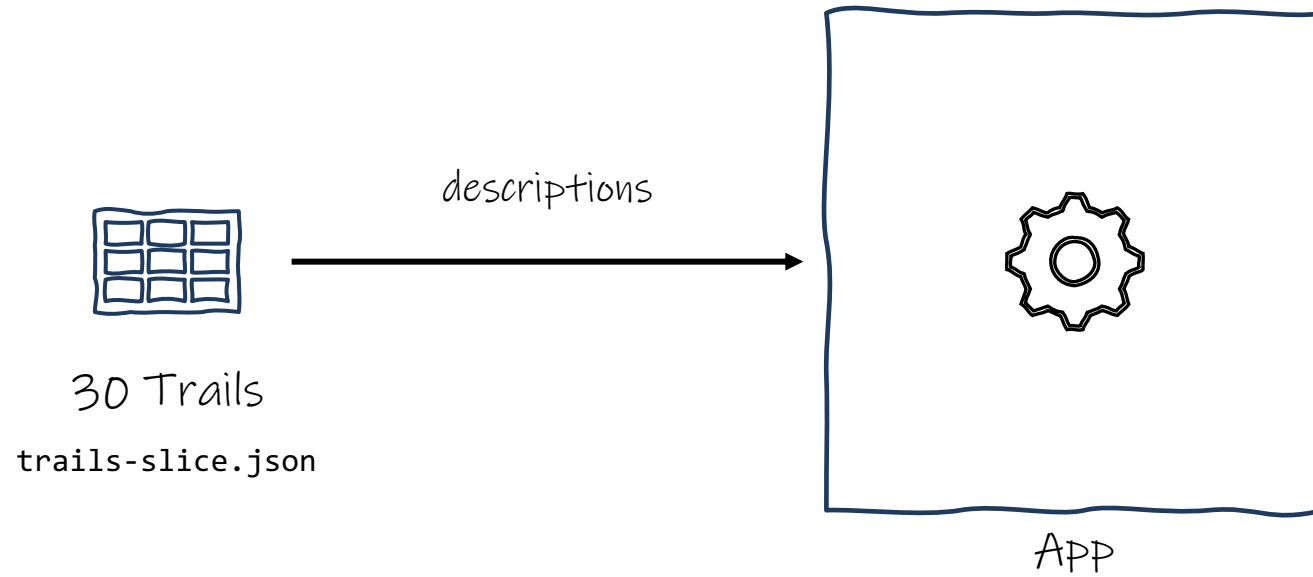


3D Trails

`trails-slice.json`

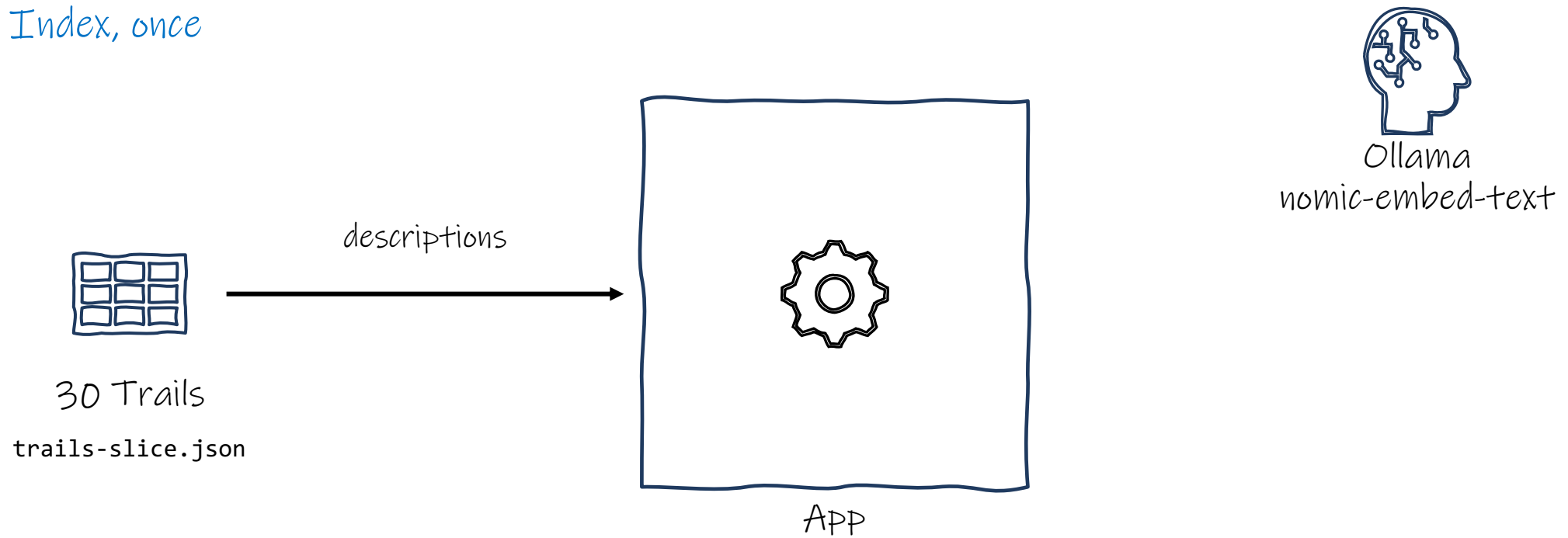
F04 · How It Works

1 · Index, once



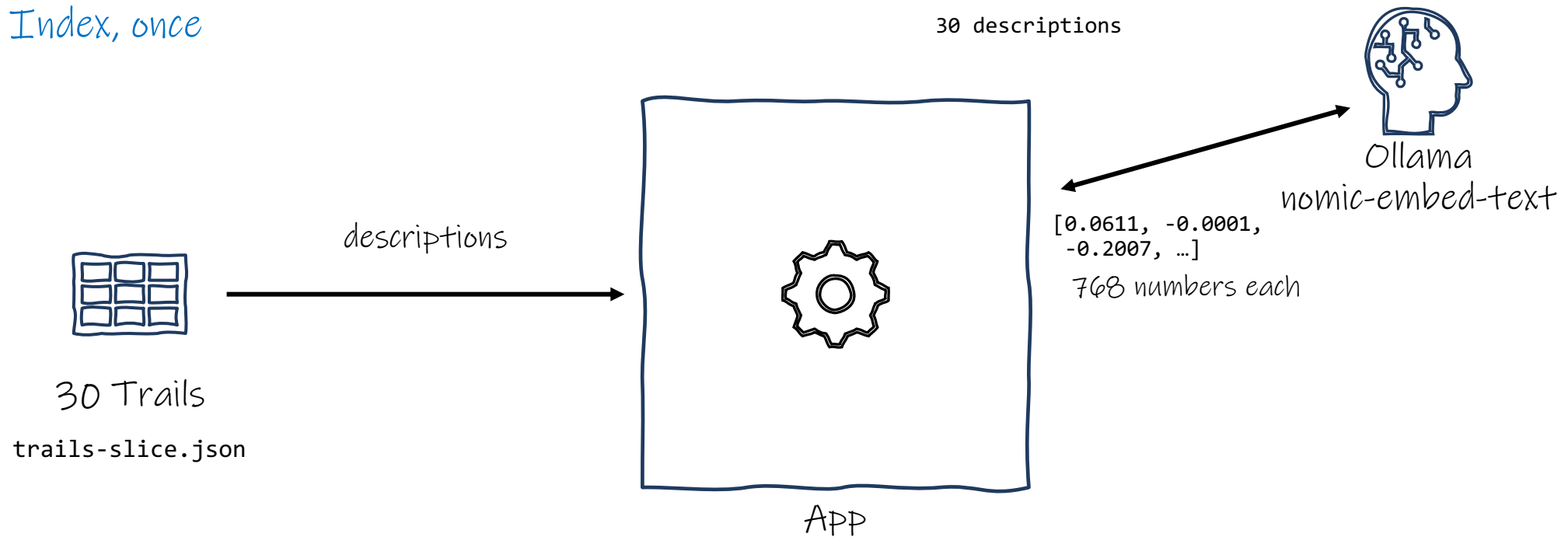
F04 · How It Works

1 · Index, once



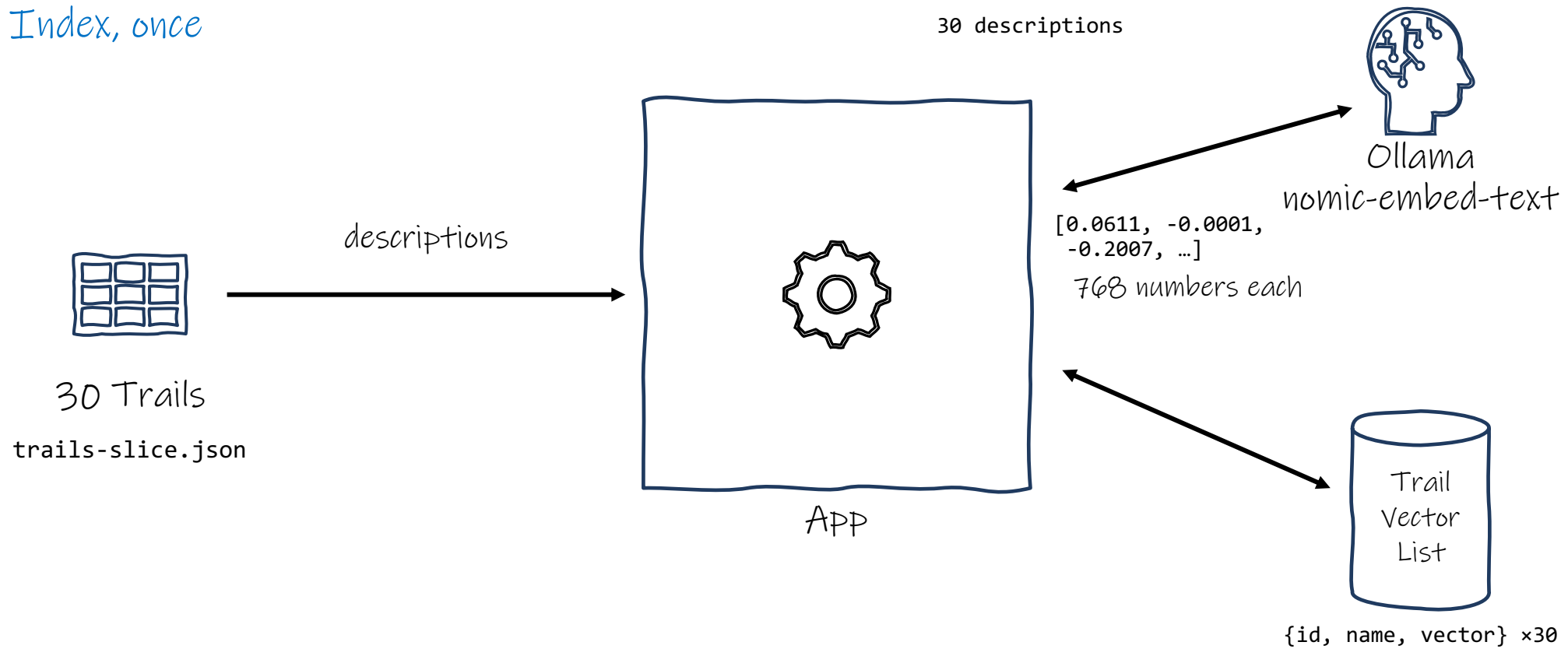
F04 · How It Works

1 · Index, once



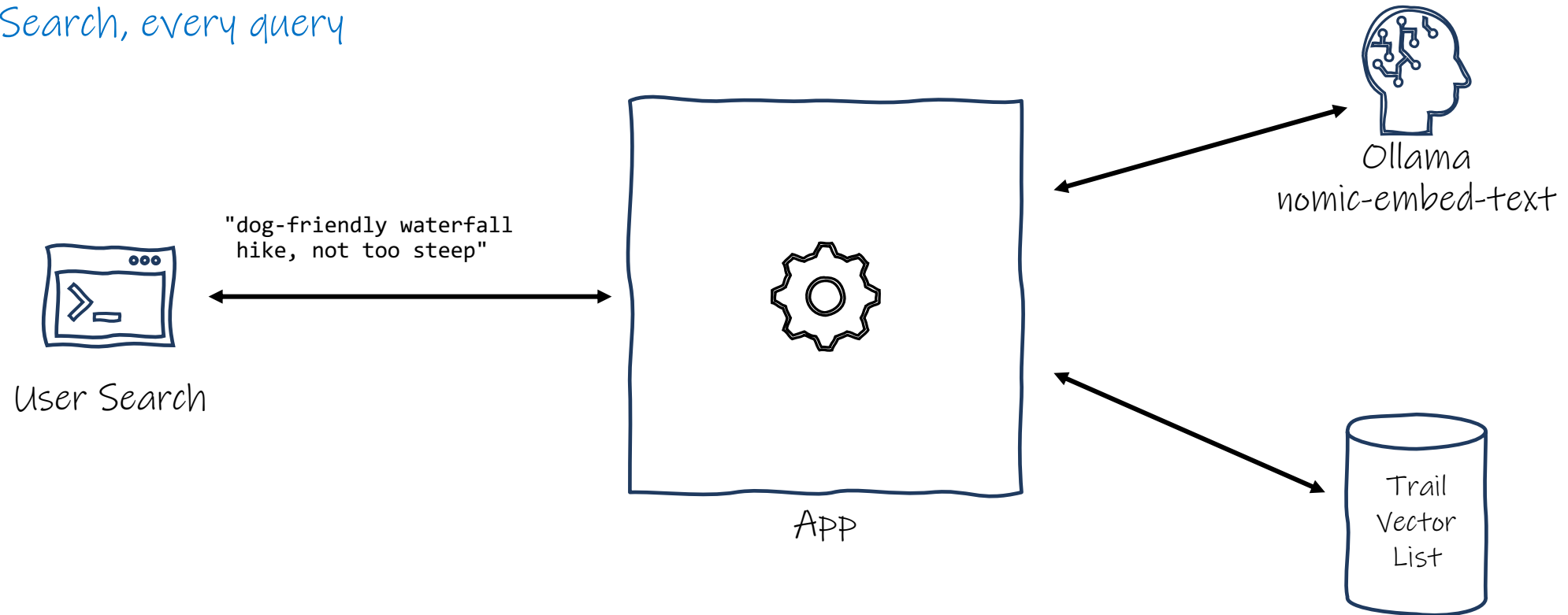
F04 · How It Works

1 · Index, once



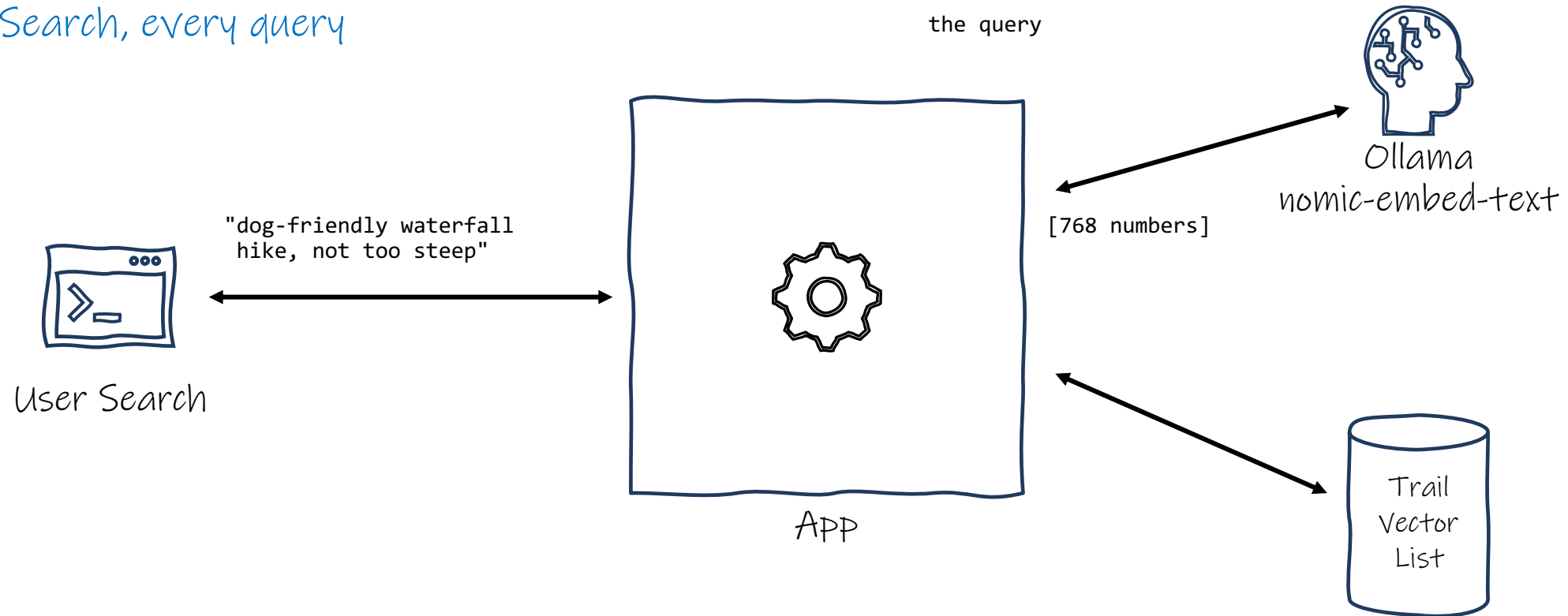
F04 · How It Works

2 · Search, every query



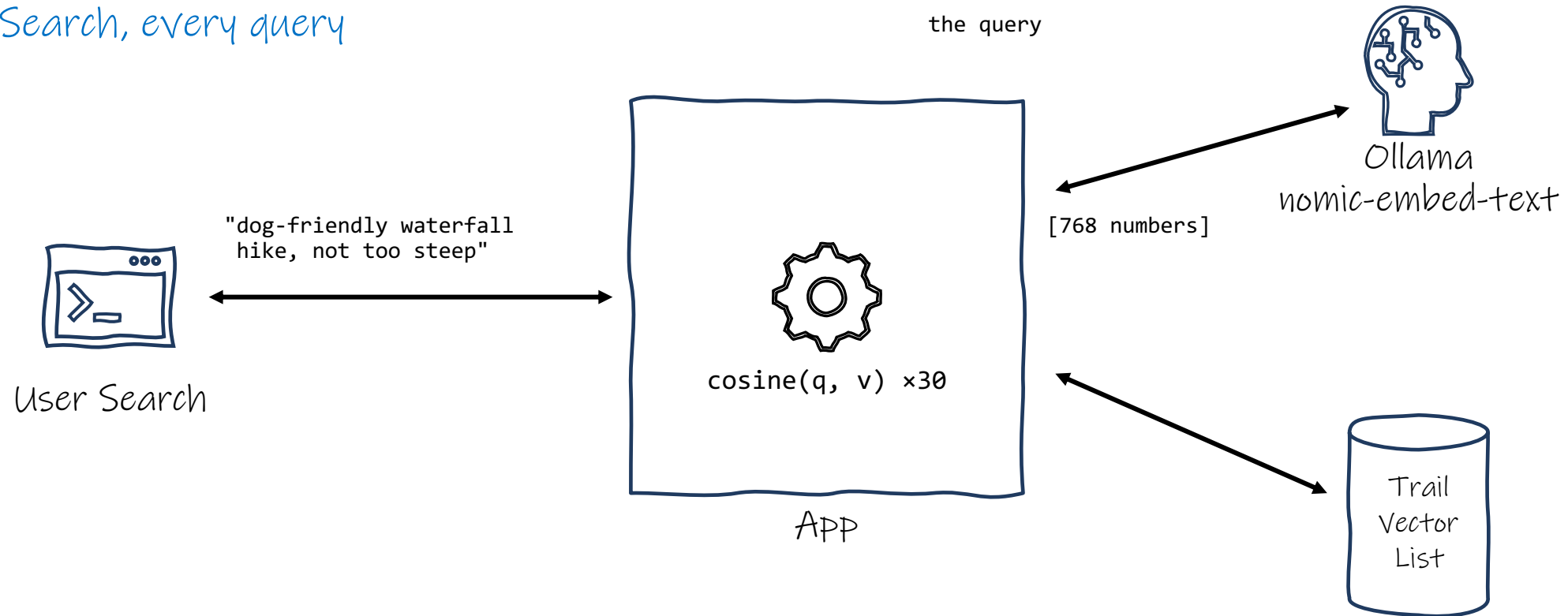
F04 · How It Works

2 · Search, every query



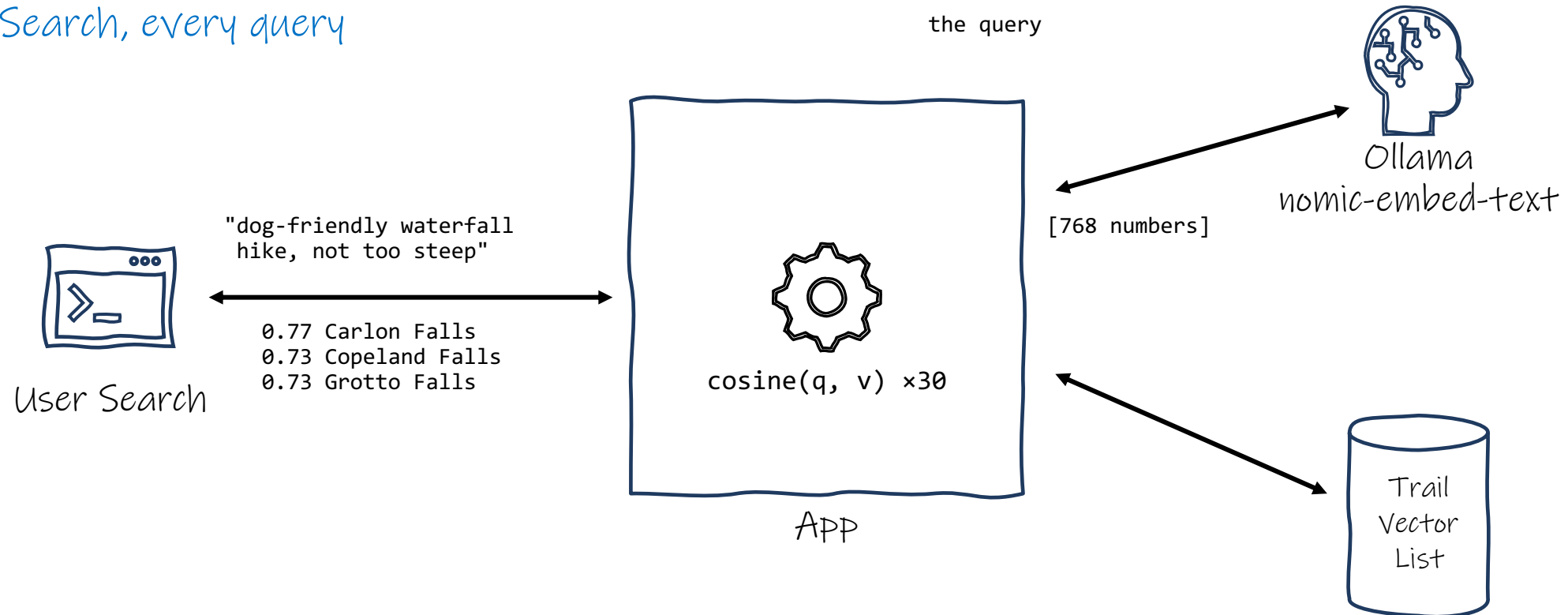
F04 · How It Works

2 · Search, every query



F04 · How It Works

2 · Search, every query



F05 · RAG



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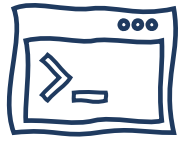


Look in your documents as the source; the model **composes an answer from what it finds.**



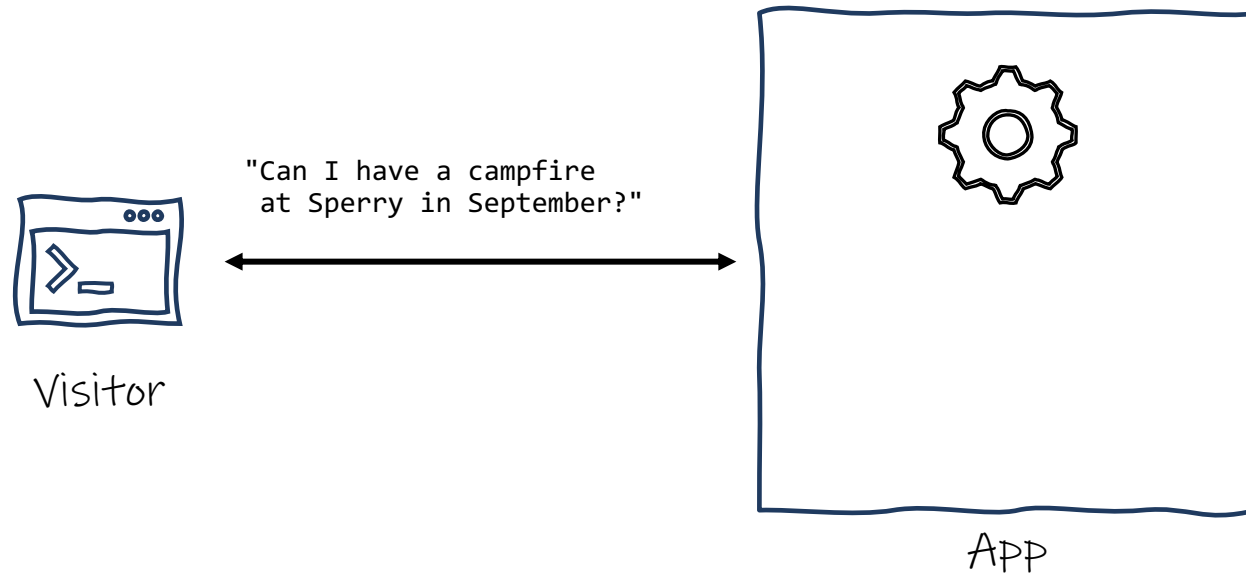
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F05 · How It Works

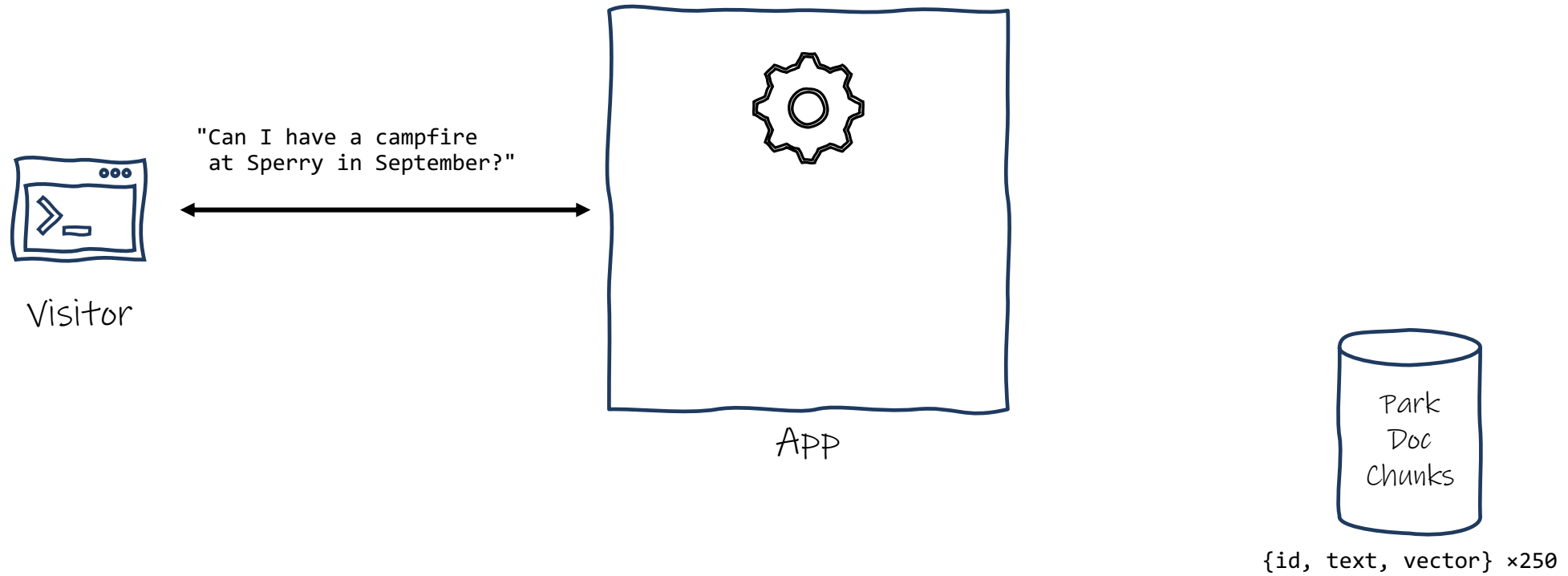


Visitor

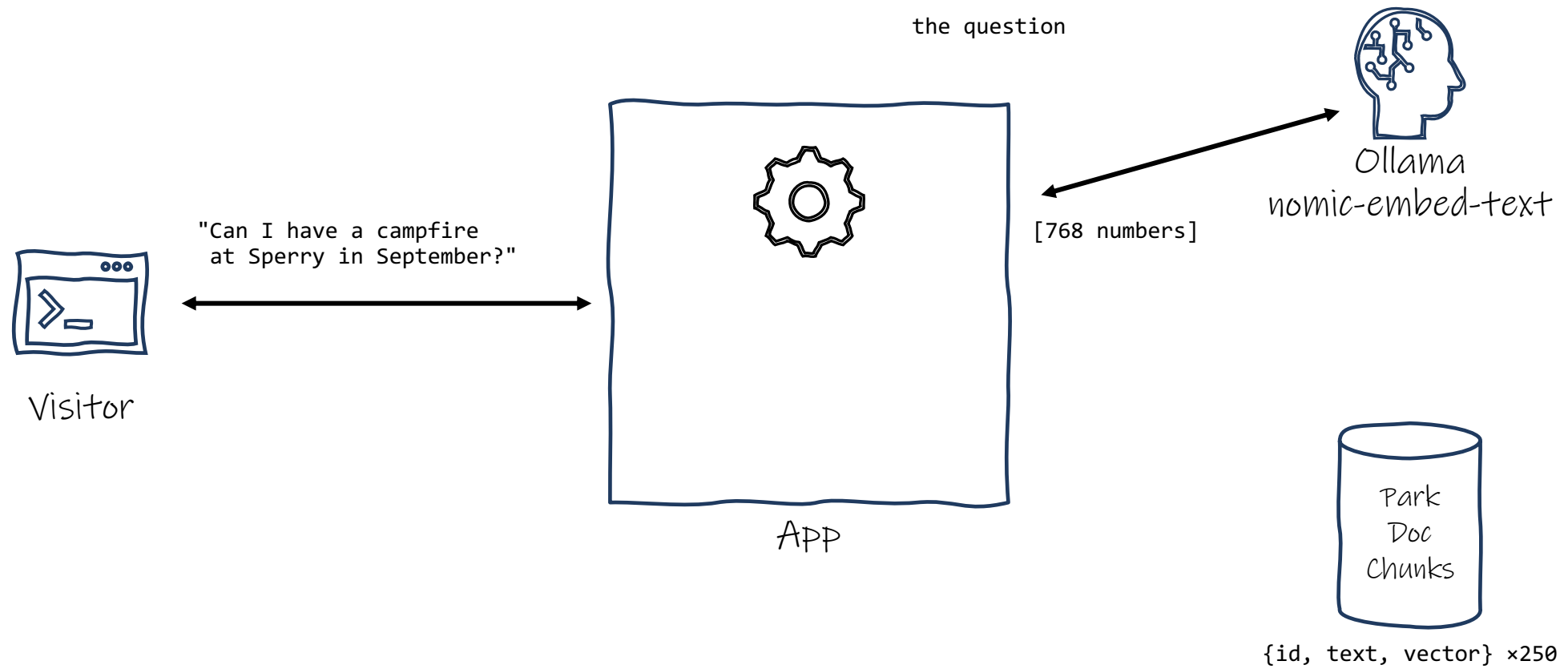
F05 · How It Works



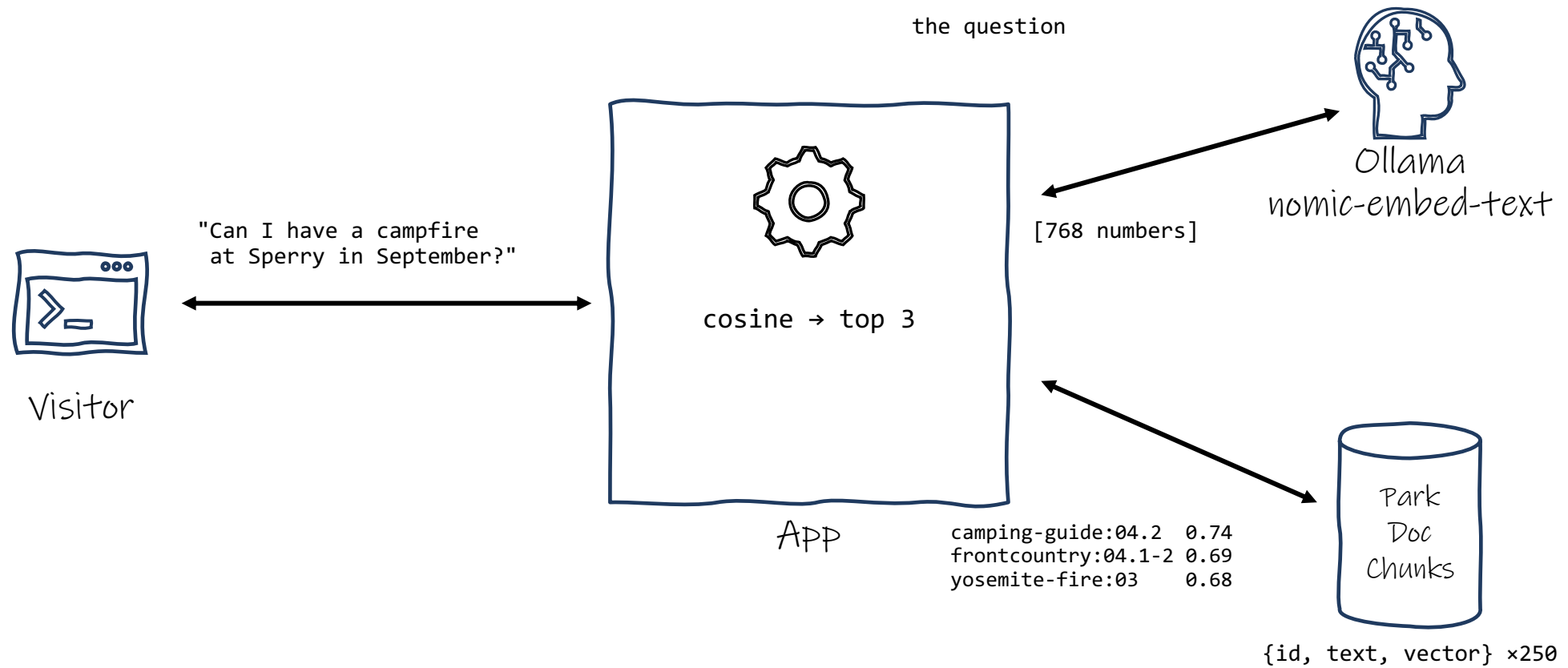
F05 · How It Works



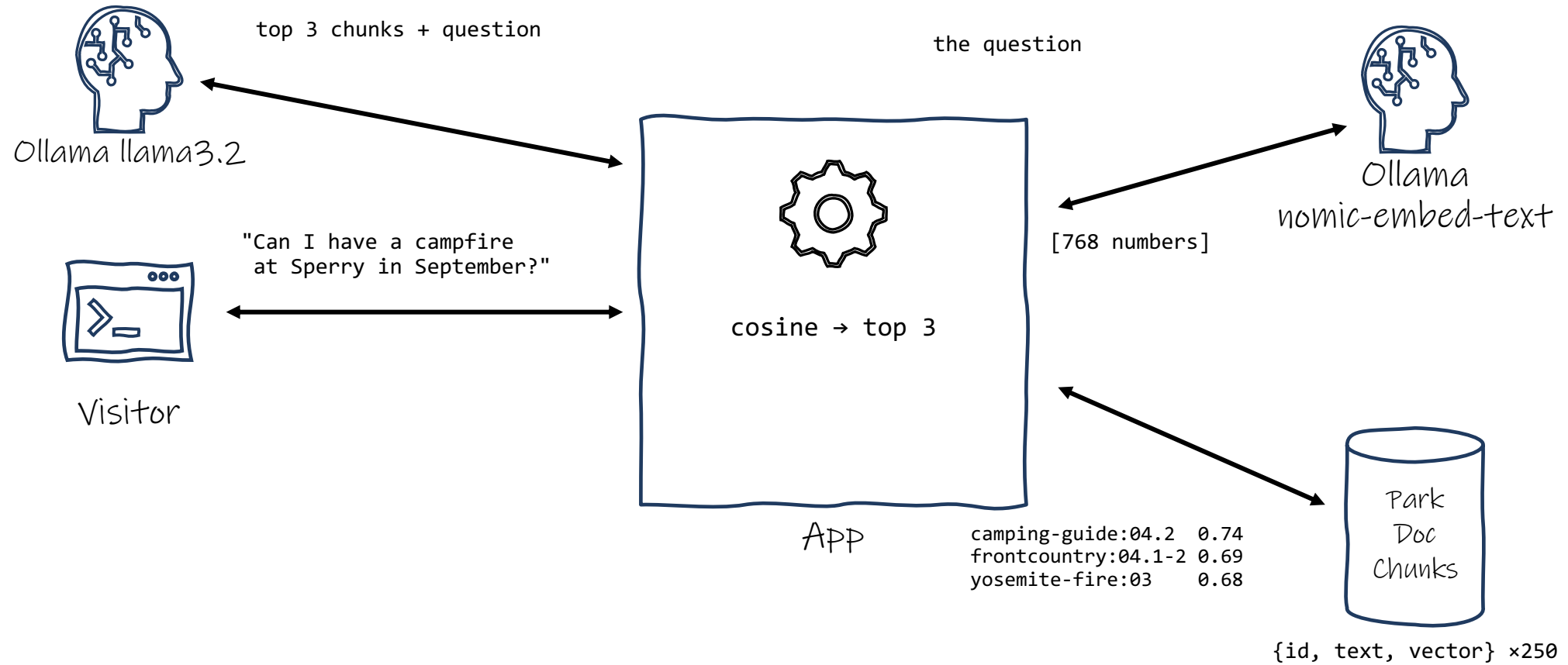
F05 · How It Works



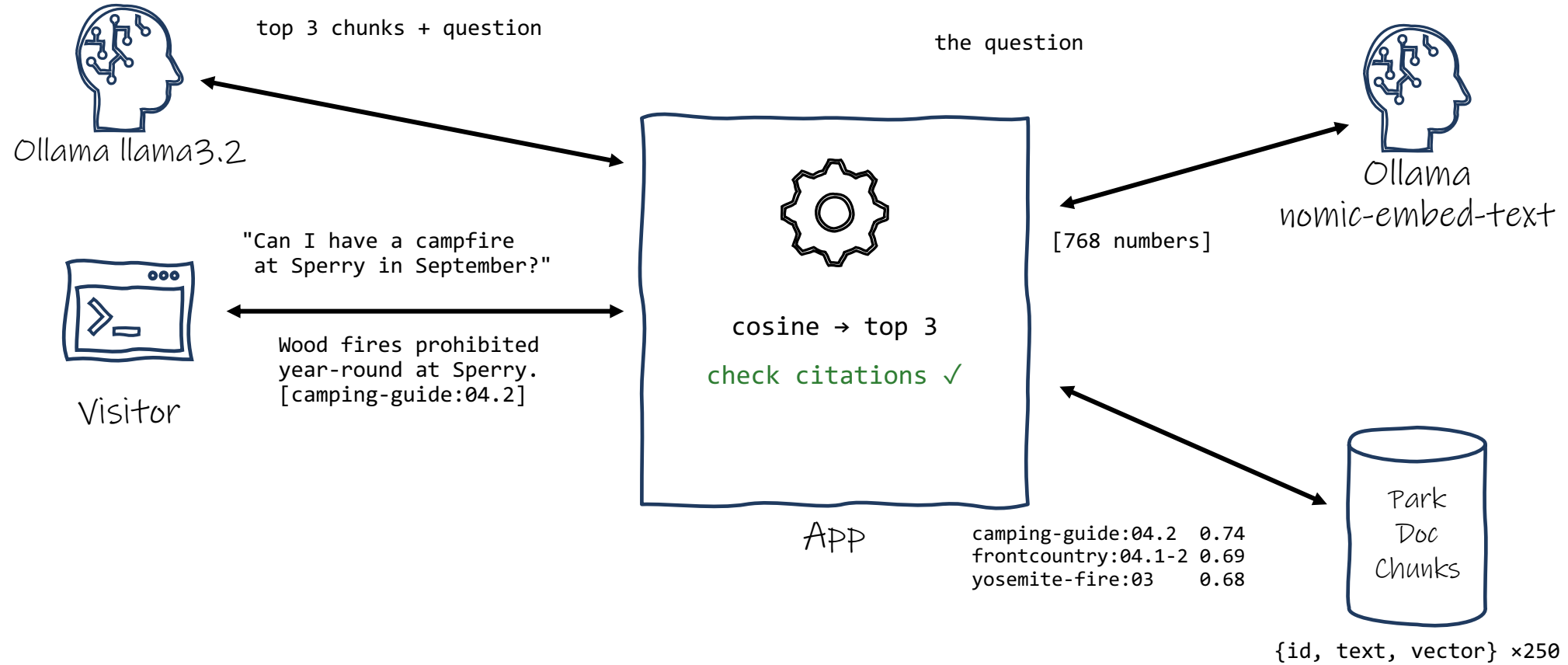
F05 · How It Works



F05 · How It Works



F05 · How It Works



F05 · Before, and Retrieve

```
if (question == "") question = "Is the Avalanche Lake Trail open right now?";  
IChatClient llm = Clients.Cloud("gpt-4.1");  
  
// Before: no documents. The model answers from memory.  
if (!grounded) { await Stream(llm, question); return; }  
  
// 1. Retrieve: the three chunks closest to the question.  
var top = await Retrieve(question, topK: 3);
```

F05 · Before, and Retrieve

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if (question == "") question = "Is the Avalanche Lake Trail open right now?";
IChatClient llm = Clients.Cloud("gpt-4.1");

// Before: no documents. The model answers from memory.
if (!grounded) { await Stream(llm, question); return; }

// 1. Retrieve: the three chunks closest to the question.
var top = await Retrieve(question, topK: 3);
```

F05 · Ground

```
var answer = await Stream(llm, $""
```

You are a park information assistant. Answer the visitor's question using ONLY the context below.

- Do not use outside knowledge.
- Cite the `chunk_id` of each chunk you relied on, in square brackets, exactly as it appears in the context.
- If none of the context is relevant, reply exactly: "{Refusal}"

A question about "right now" is answered from the context:
today is September 23, 2026, and a notice in effect "until further notice" is still in effect.

Context:

```
{string.Join("\n\n", top.Select(c => $"chunk_id: {c.Id}\n{c.Text}"))}
```

Question: {question}

```
""");
```

F05 · Ground

```
var answer = await Stream(llm, $""  
  You are a park information assistant. Answer the visitor's  
  question using ONLY the context below.  
  - Do not use outside knowledge.  
  - Cite the chunk_id of each chunk you relied on, in square  
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  A question about "right now" is answered from the context:  
  today is September 23, 2026, and a notice in effect "until  
  further notice" is still in effect.
```

Context:

```
{string.Join("\n\n", top.Select(c => $"chunk_id: {c.Id}\n{c.Text}"))}
```

Question: {question}

```
""");
```

F05 · Ground

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var answer = await Stream(llm, """
You are a park information assistant. Answer the visitor's
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- Do not use outside knowledge.
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A question about "right now" is answered from the context:
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```

Context:

```
{string.Join("\n\n", top.Select(c => $"chunk_id: {c.Id}\n{c.Text}"))}
```

Question: {question}

```
""");
```

F05 · Verify

```
// 3. Verify: every citation must point at a chunk we retrieved.
var bad = Citations(answer).Where(id => !top.Any(c => c.Id == id));
Console.WriteLine(bad.Any()
    ? $"\\n!! INVALID CITATIONS: {string.Join(", ", bad)}"
    : $"\\n[citations valid: {string.Join(", ", Citations(answer).Distinct())}]");
```

DEMO · F05 RAG

F06 · Recommendations



TRAILHEAD
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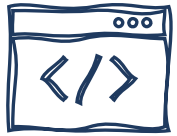


Suggest the **next thing** a user will want,
from what they liked already.



TRAILHEAD
TECHNOLOGY PARTNERS

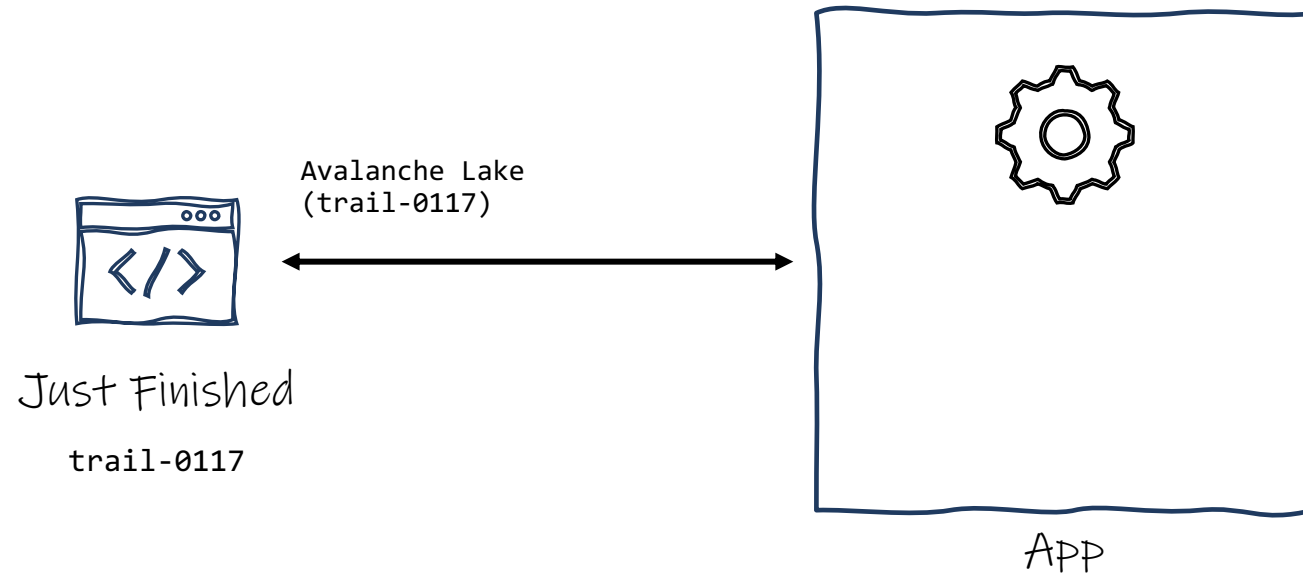
F06 · How It Works



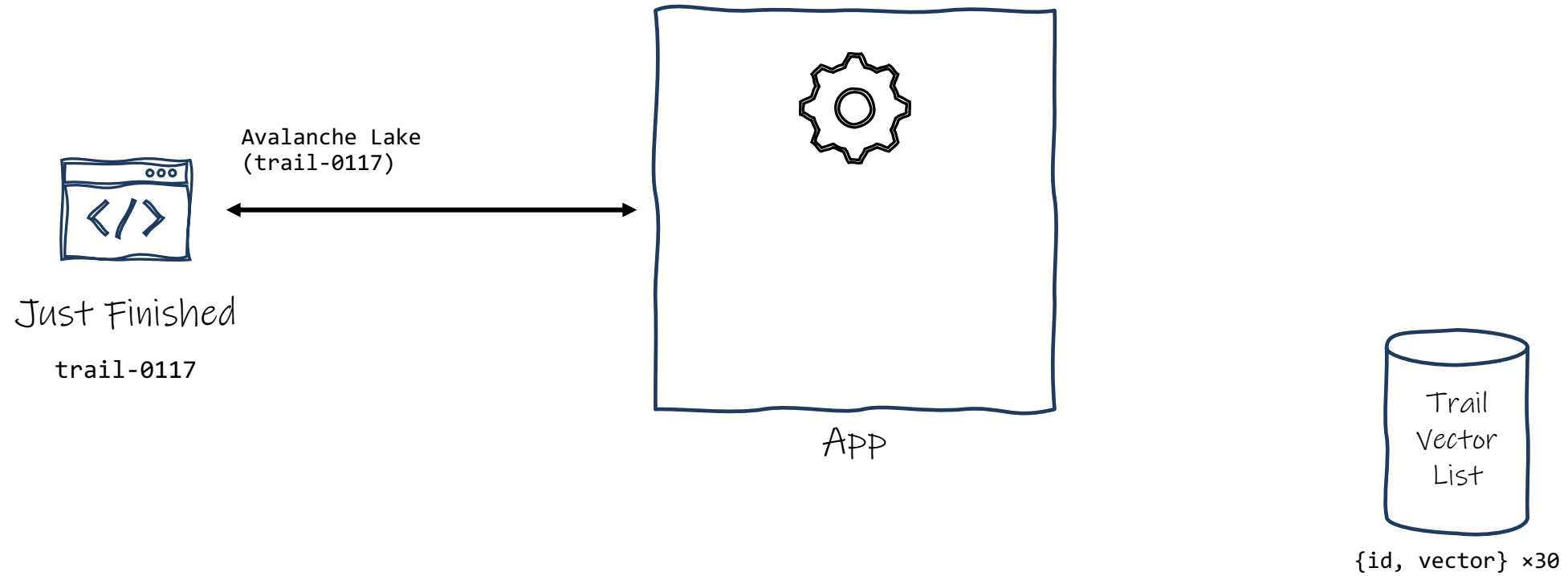
Just Finished

trail-0117

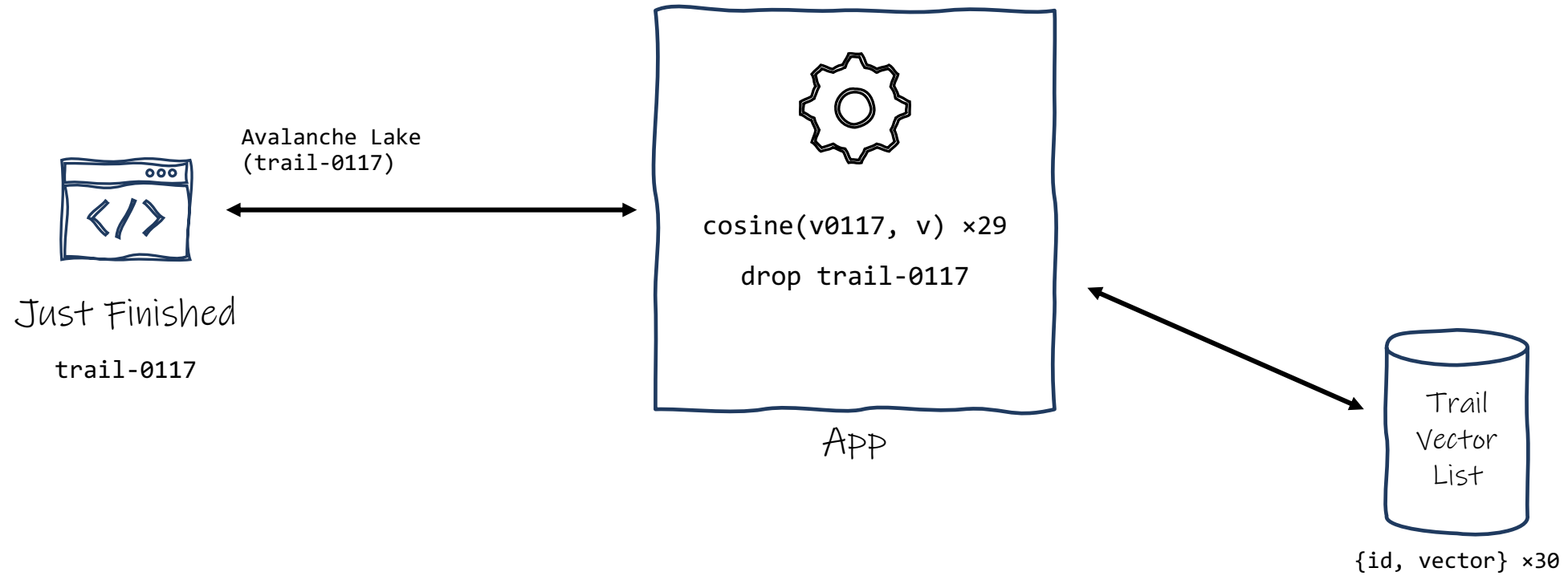
F06 · How It Works



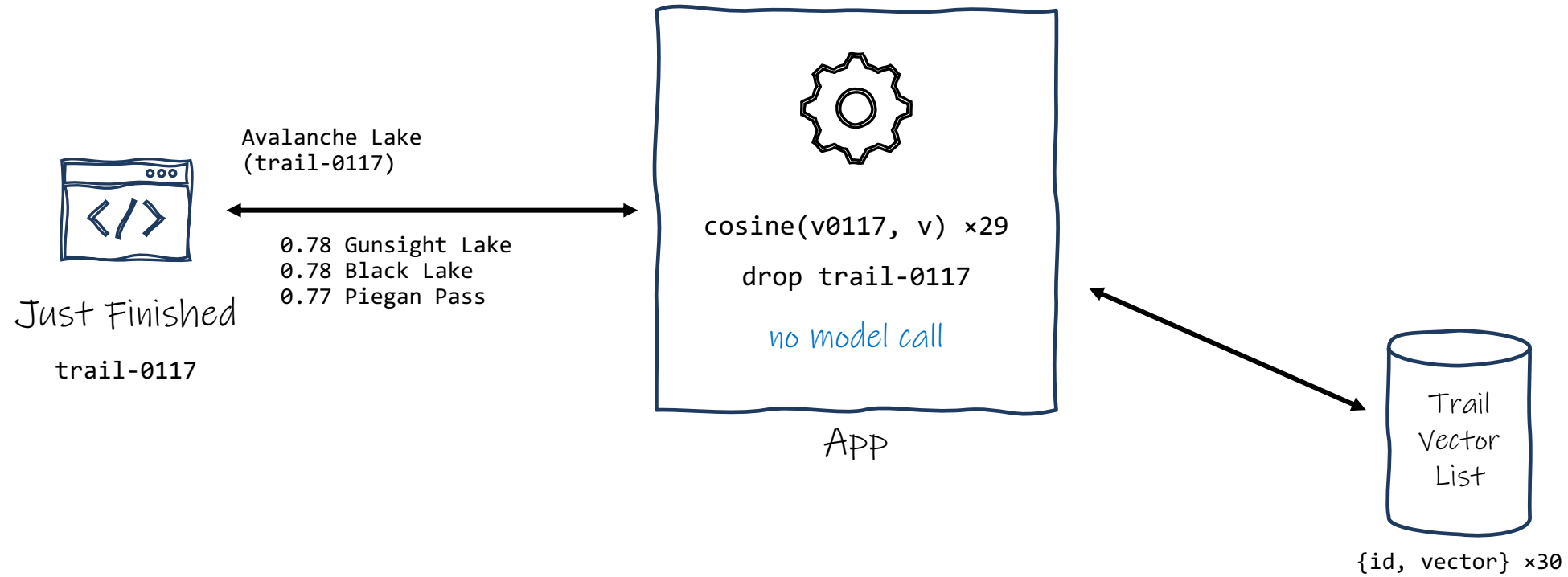
F06 · How It Works



F06 · How It Works



F06 · How It Works



F07 · Classification & Routing



TRAILHEAD
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Give each message a **label**, and let the label pick the **queue**.



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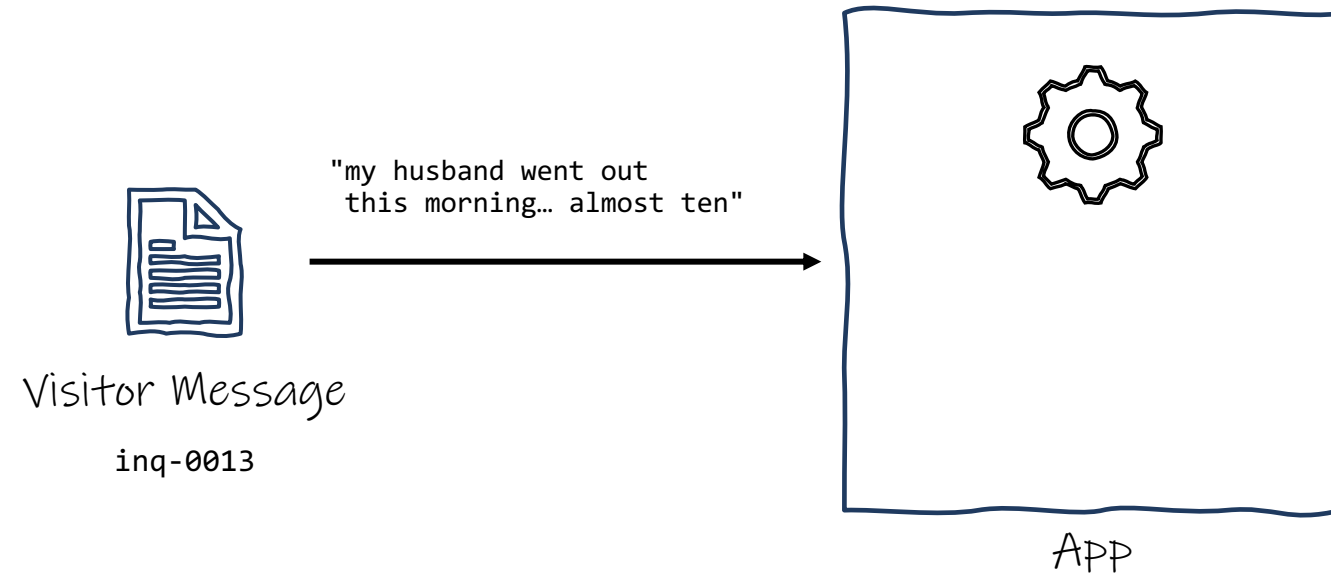
F07 · How It Works



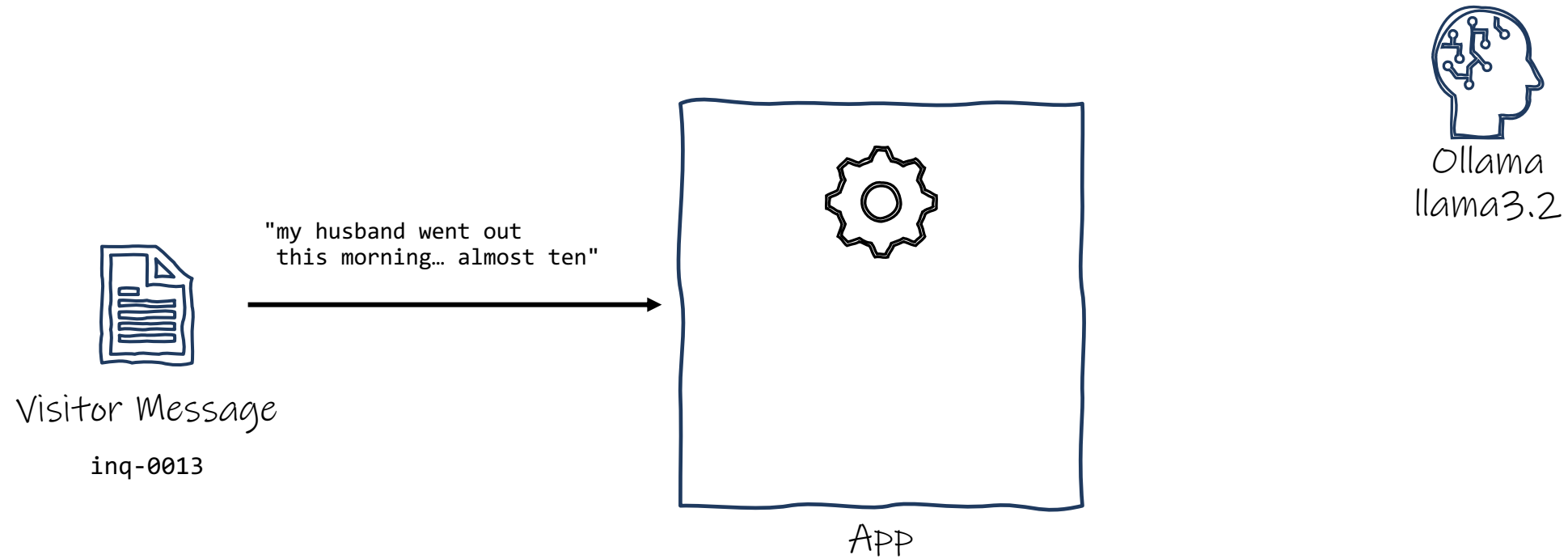
Visitor Message

inq-0013

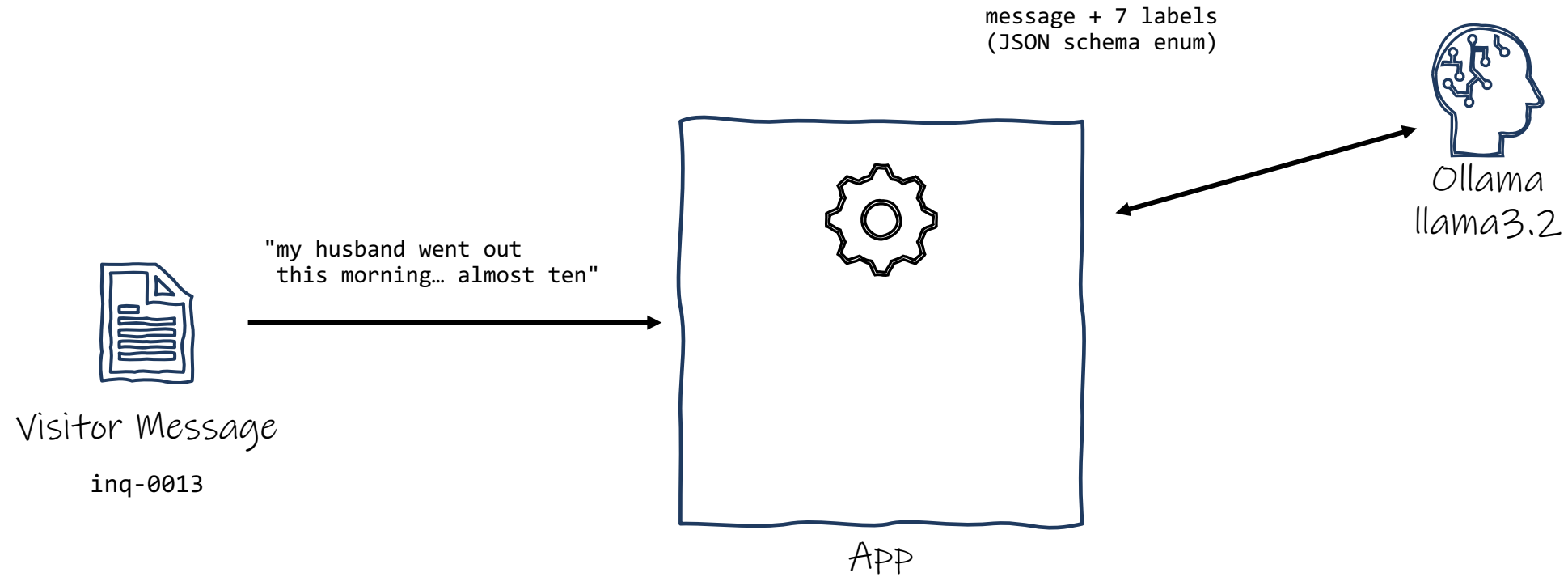
F07 · How It Works



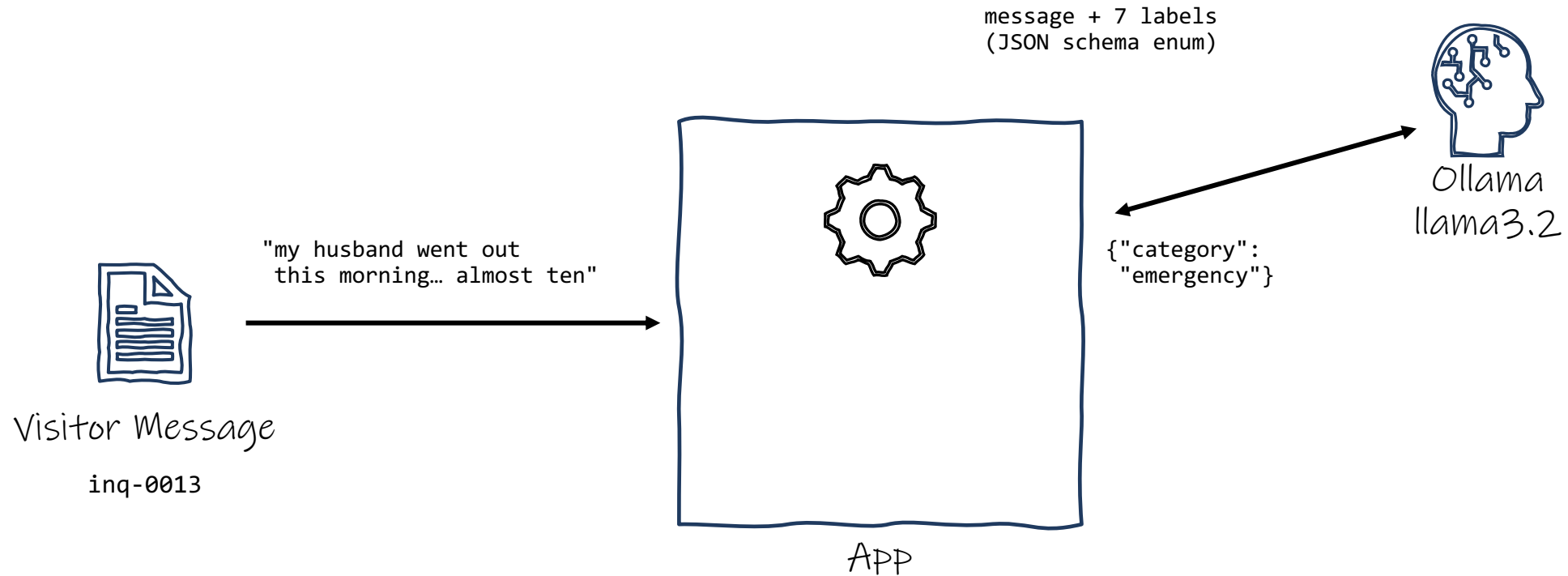
F07 · How It Works



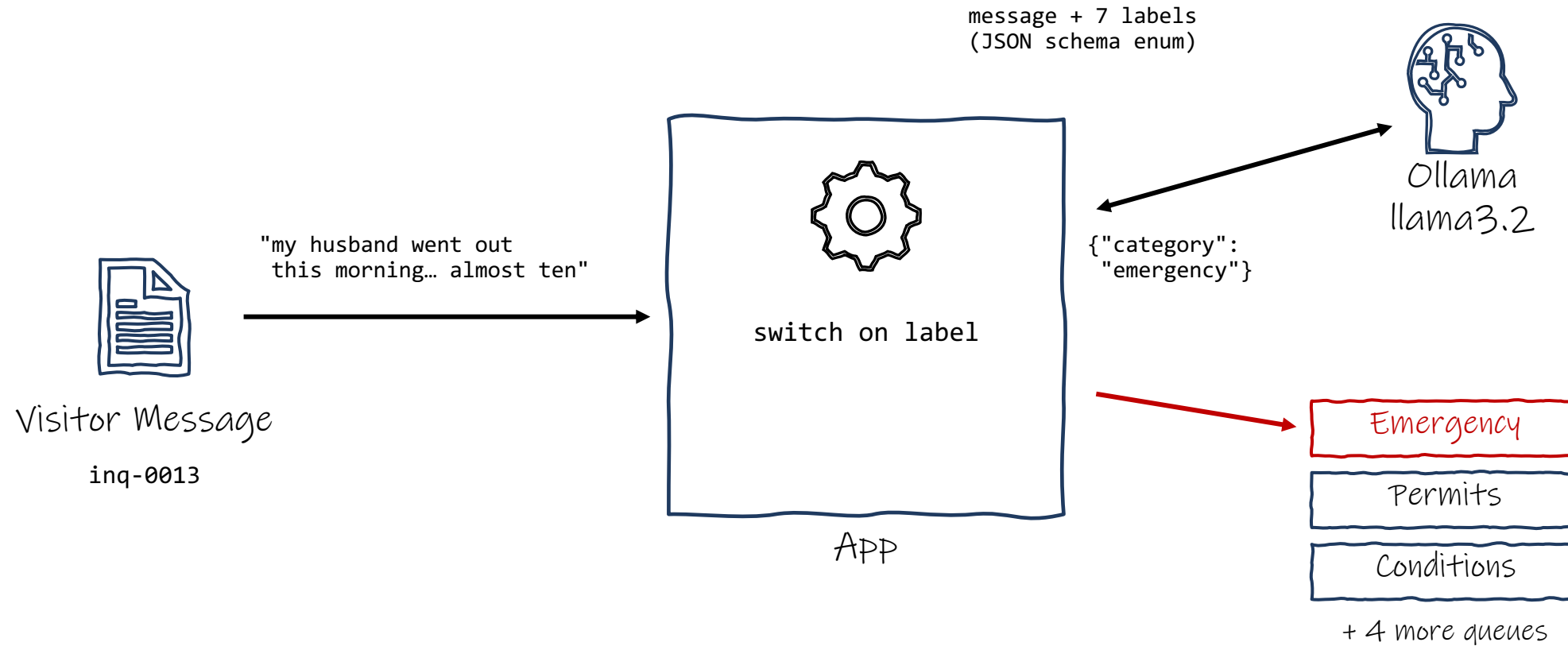
F07 · How It Works



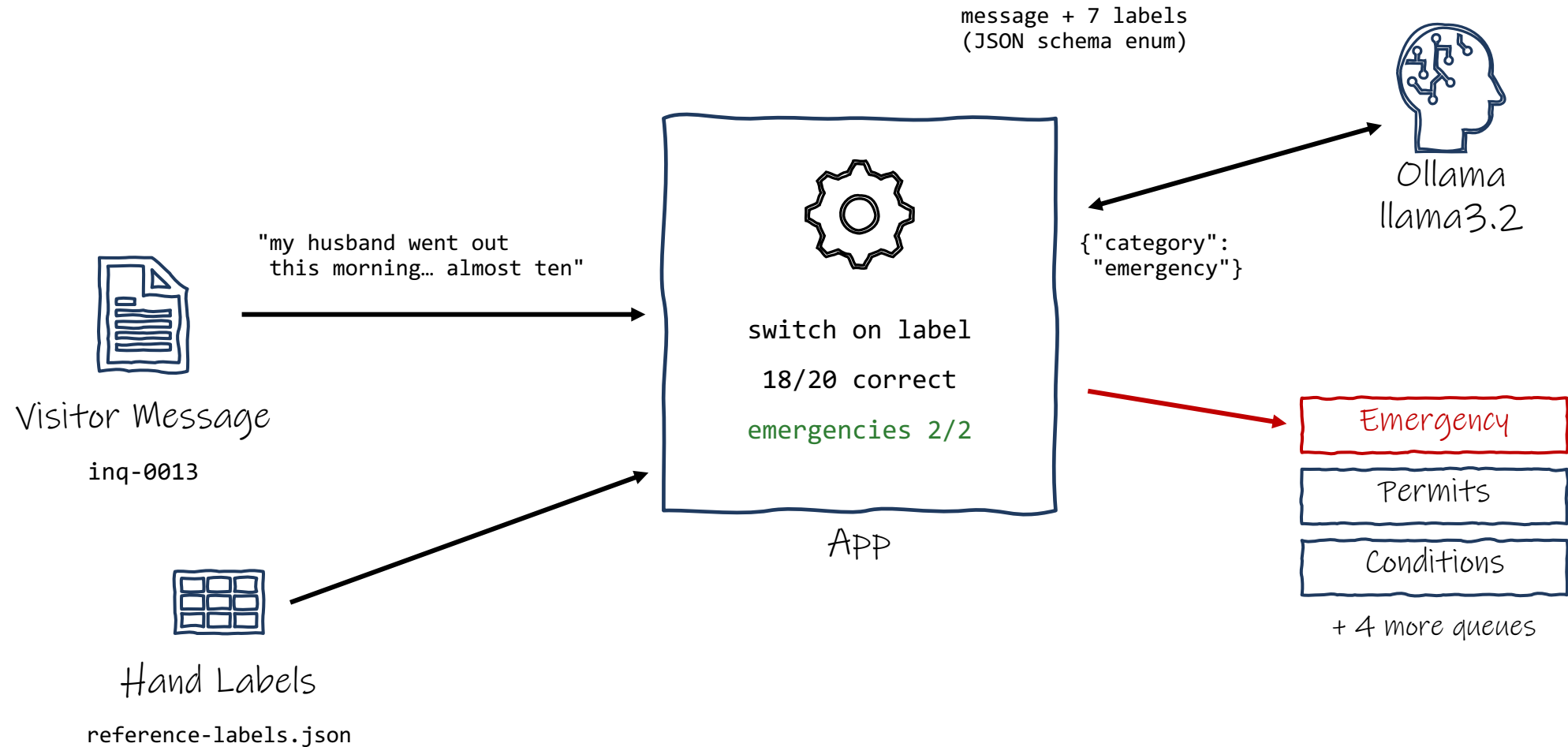
F07 · How It Works



F07 · How It Works



F07 · How It Works



F07 · The Loop

```
IChatClient client = Clients.Local("llama3.2");  
// Same message, same queue, every run. Scoring needs that.  
var options = new ChatOptions { Temperature = 0 };  
foreach (var inquiry in LoadInquiries())  
{  
    // The answer can only be a Category: a label with no queue  
    // behind it cannot come back.  
    var response = await client.GetResponseAsync<TriageResult>(  
        Prompt(inquiry.text), options);  
    results.Add((inquiry, response.Result.Category));  
}  
  
// Accuracy is the headline. Emergency recall decides whether it ships:  
// a missed emergency is a person waiting in a queue nobody watches.  
Score(results);
```

F07 · The Loop

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F07 · The Taxonomy Is the Prompt

- permit: reserving, changing, canceling, or paying for a permit, pass, or reservation, including billing problems and missing confirmations for a permit application.
- conditions: asking whether a trail, road, or area is open, safe, or passable right now: snow, water levels, washouts, wildlife activity, closures.
- complaint: unhappy about a park facility, service, or staff member and wants it acknowledged or fixed.
- lost-and-found: reporting a lost or found physical item.
- emergency: a person may be hurt, missing, or in danger right now and needs immediate human attention.
- general: anything else: park rules, fees, trip planning, questions that fit none of the above.
- unsure: two different queues both have to act before this message can be resolved, so no single queue owns it.

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DEMO · F07 Classification & Routing

F08 · Anomaly Detection



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Spot the item in a stream
that **doesn't look like the rest.**



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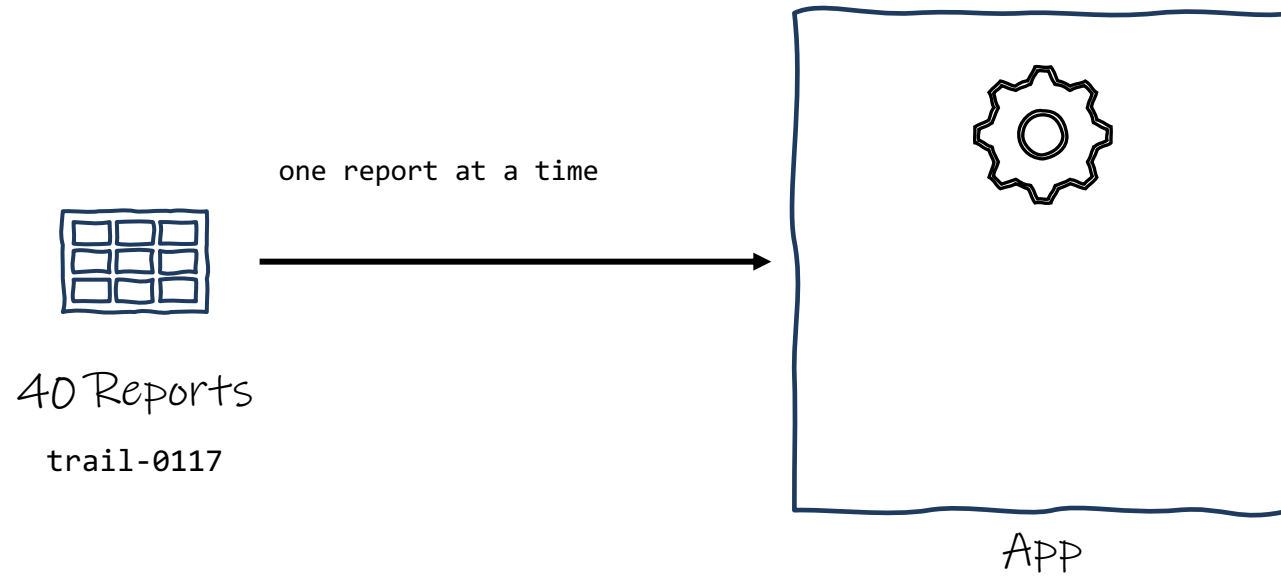
F08 · How It Works



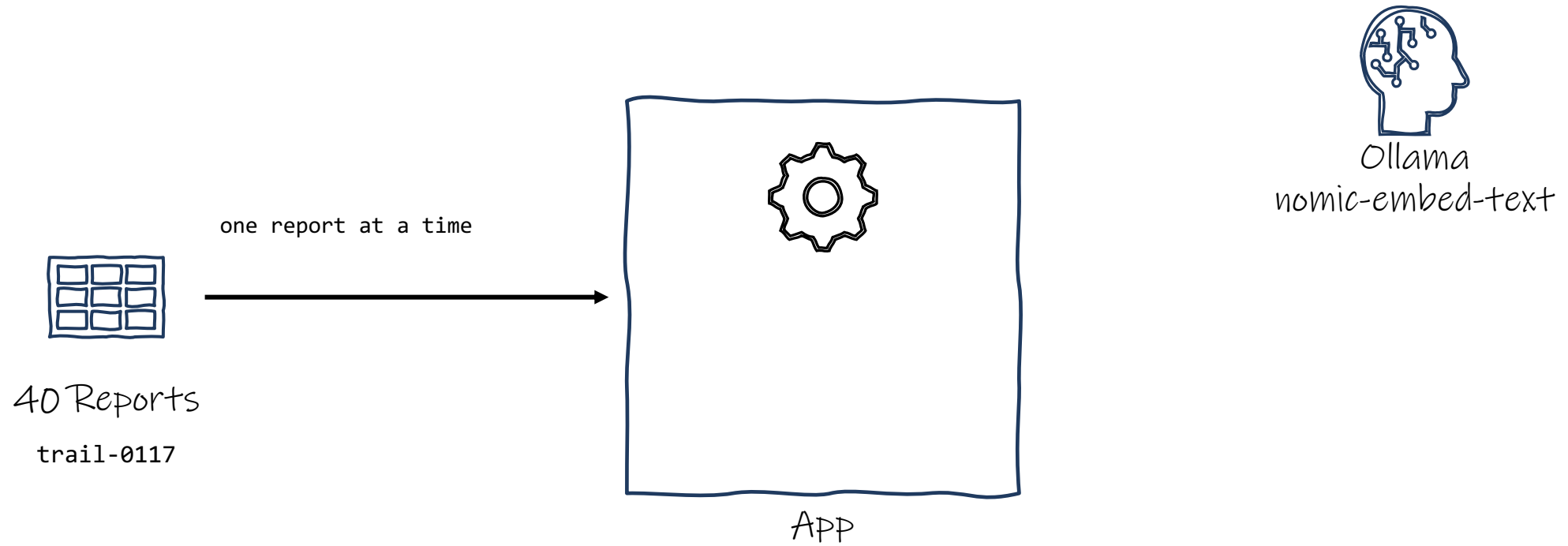
40 Reports

trail-0117

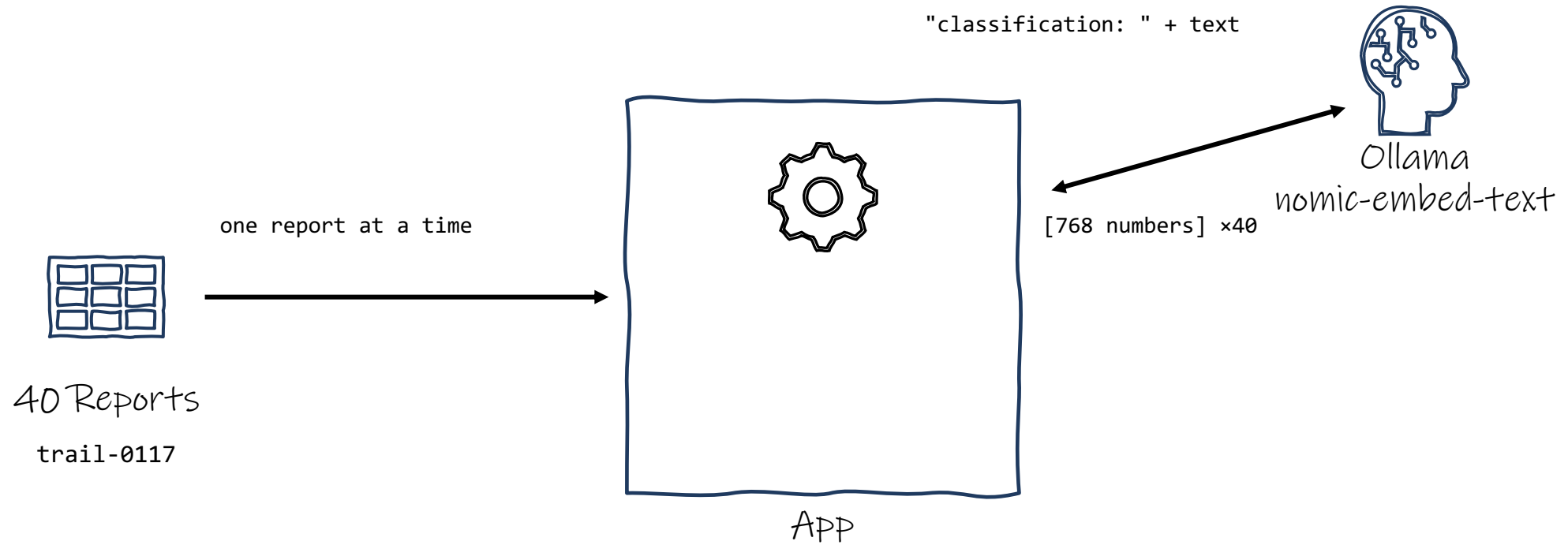
F08 · How It Works



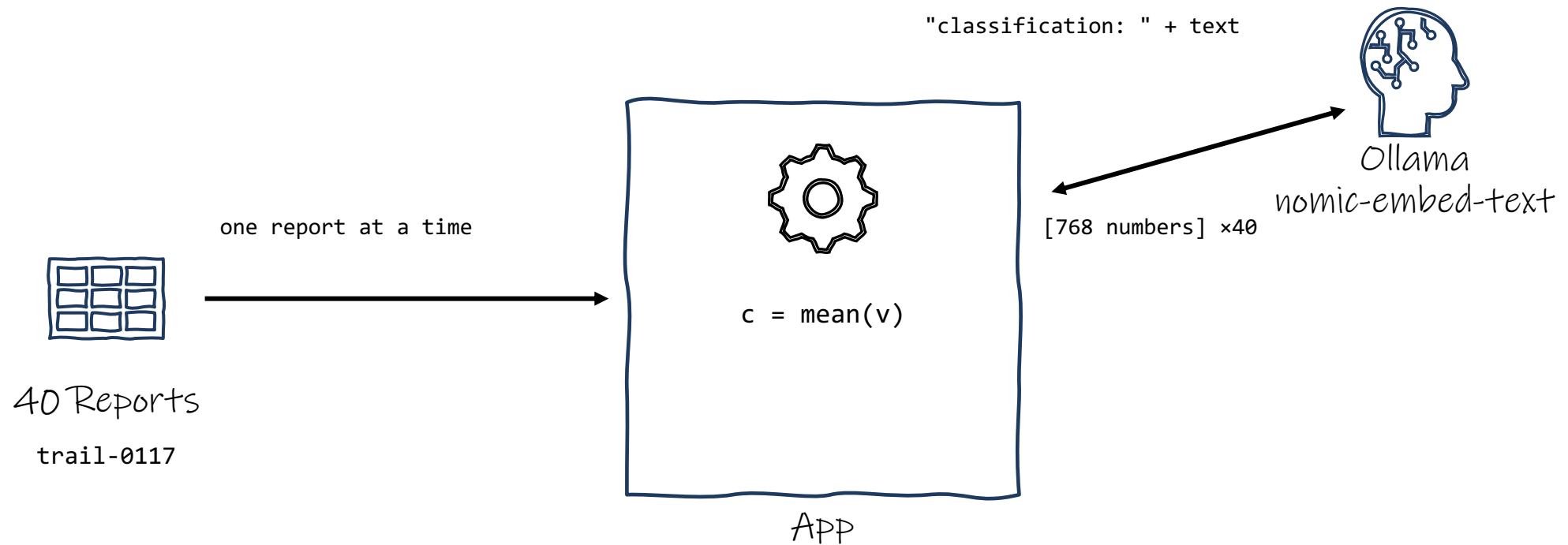
F08 · How It Works



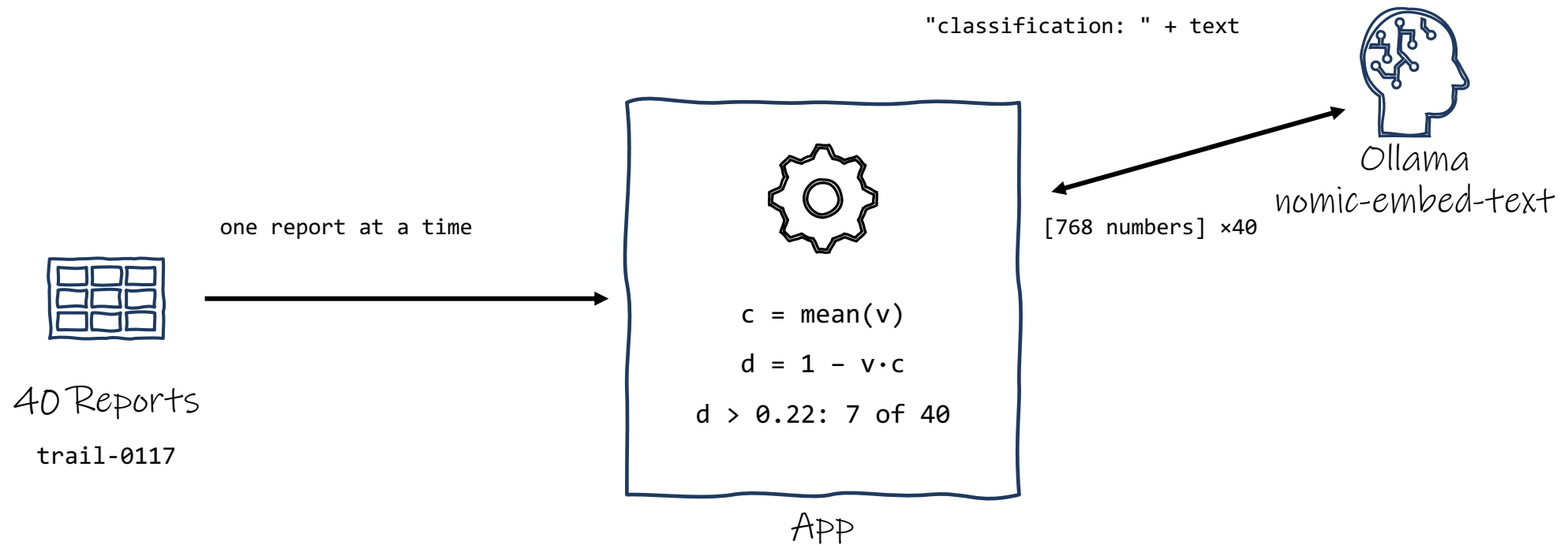
F08 · How It Works



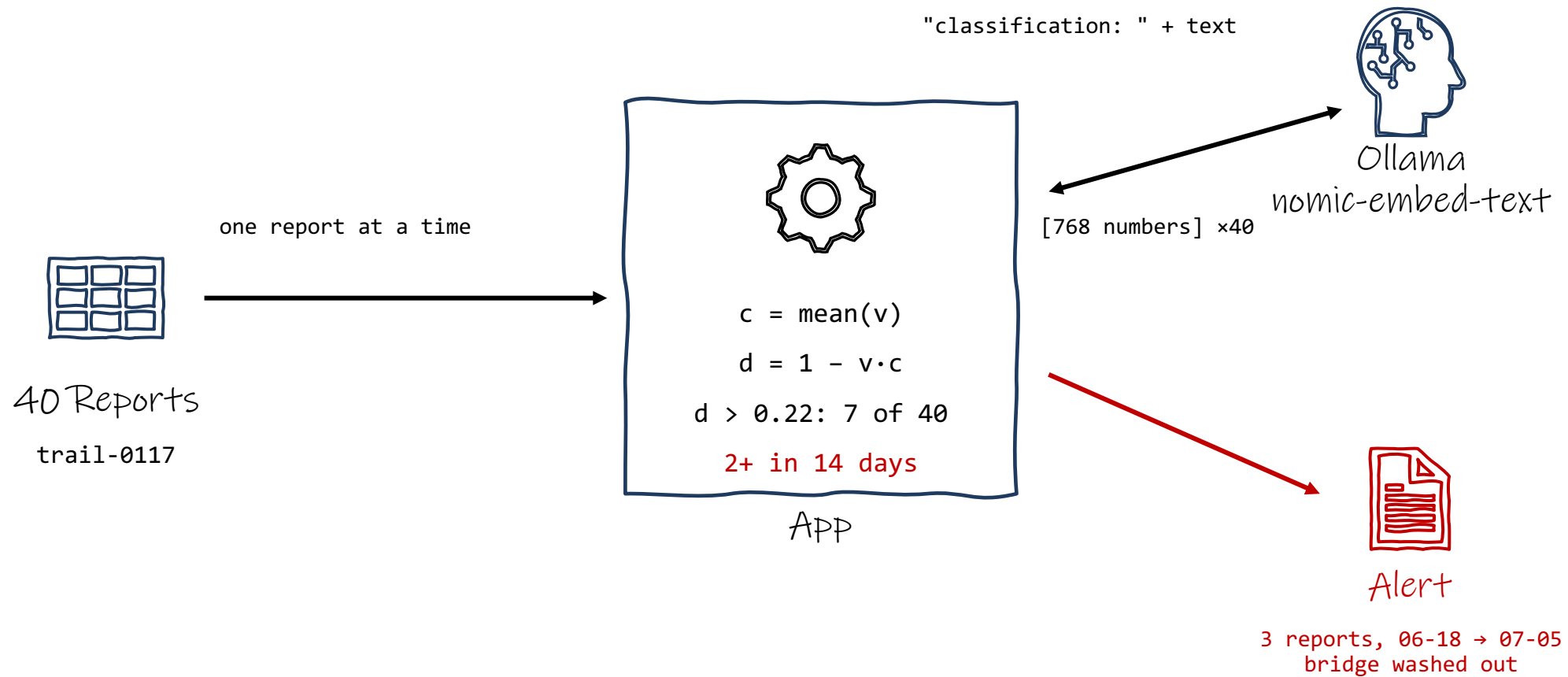
F08 · How It Works



F08 · How It Works



F08 · How It Works



F09 · Human-in-the-Loop



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The AI drafts; a **person approves** before anything goes out.



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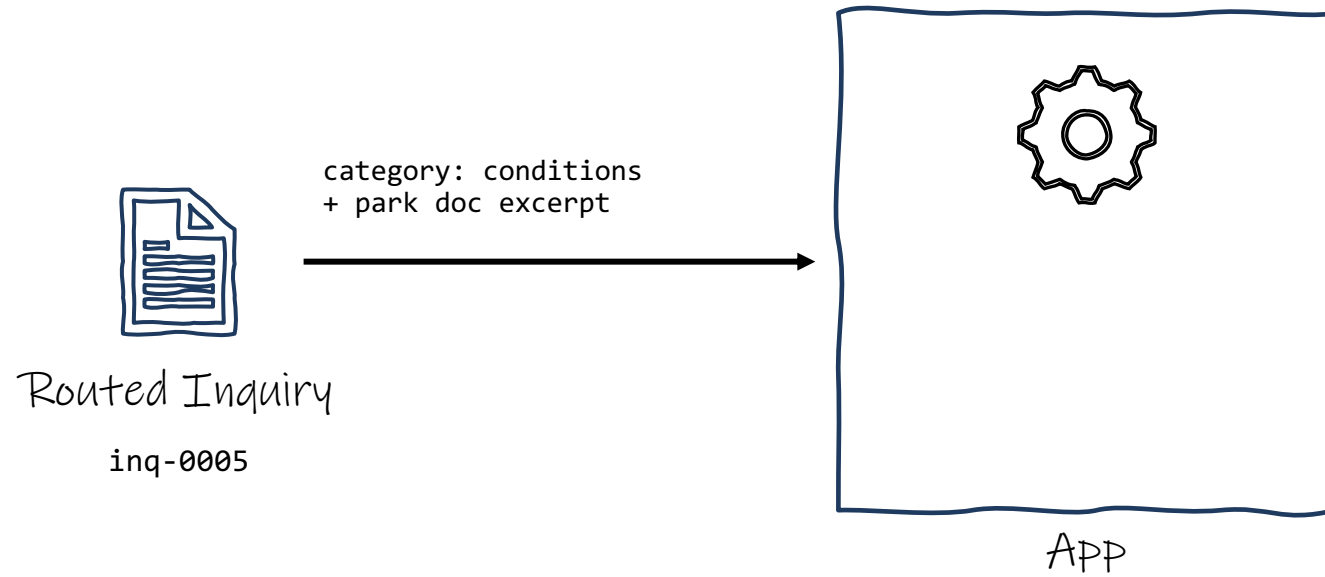
F09 · How It Works



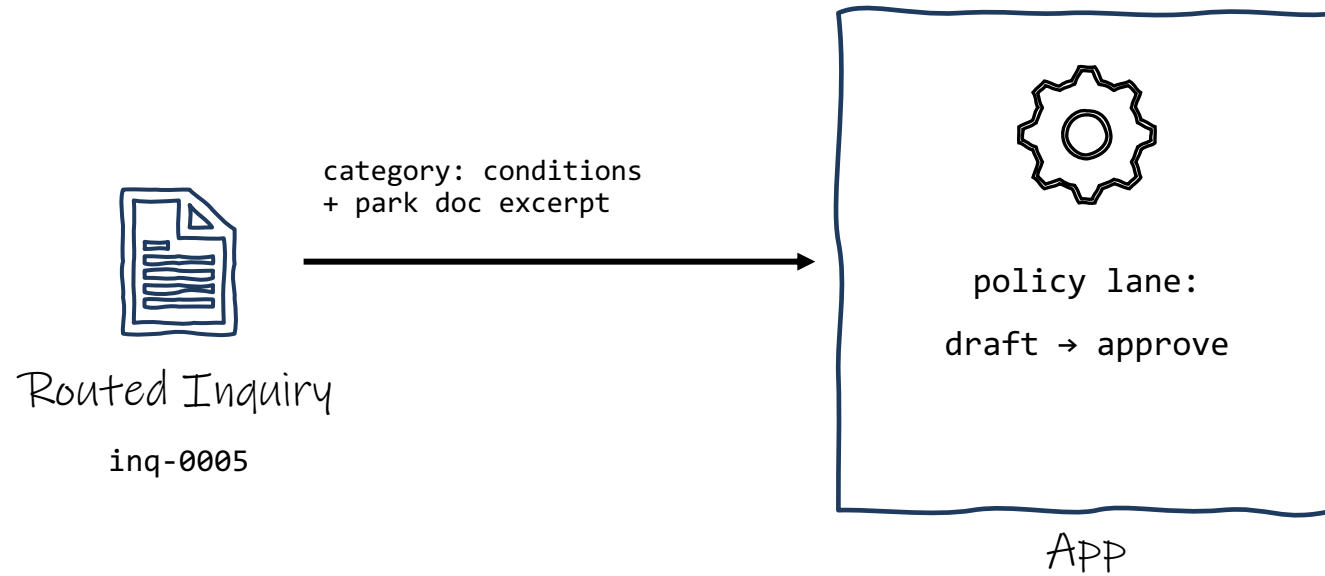
Routed Inquiry

inq-0005

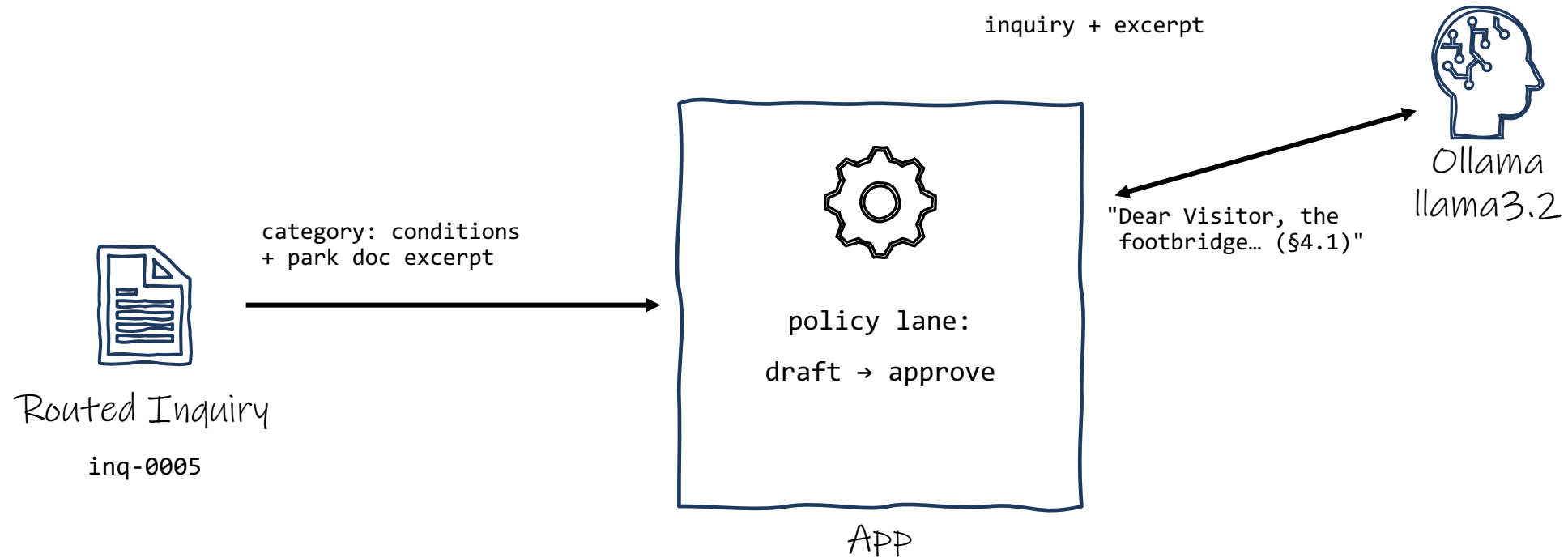
F09 · How It Works



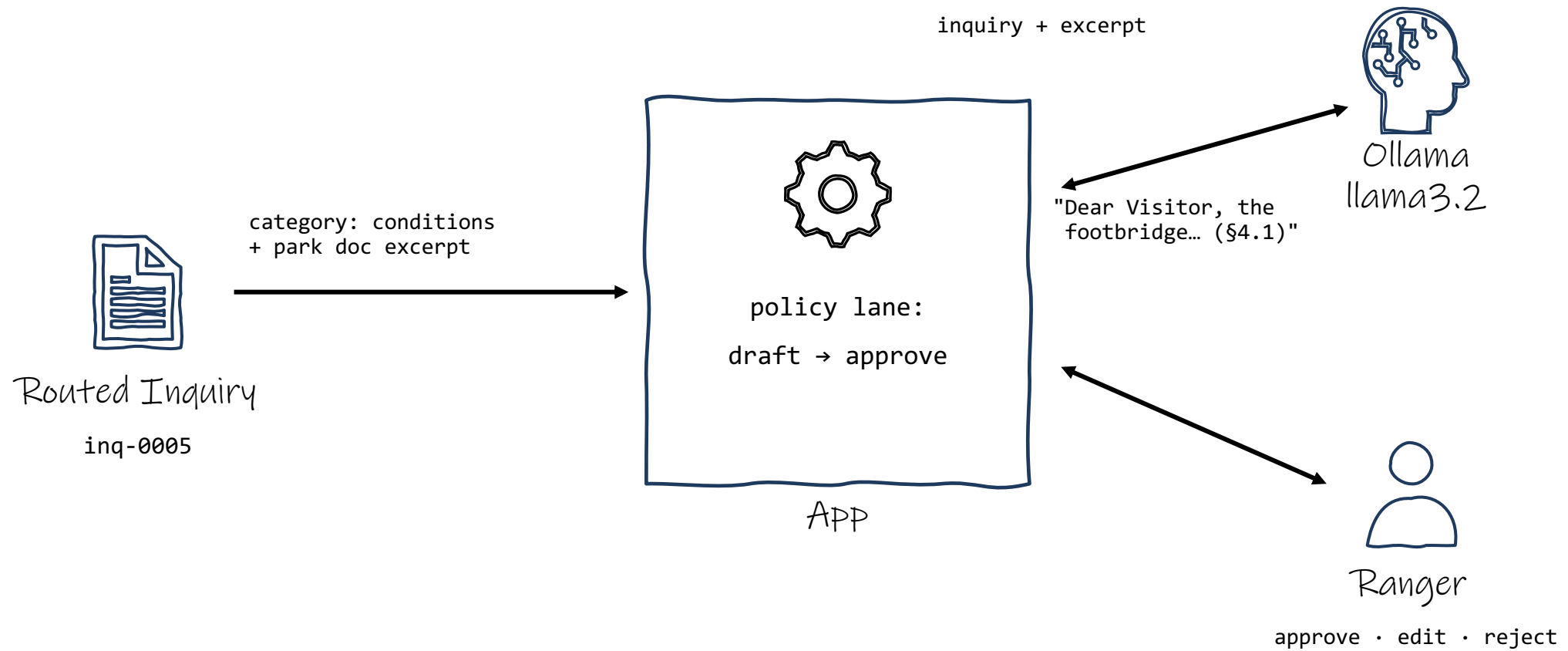
F09 · How It Works



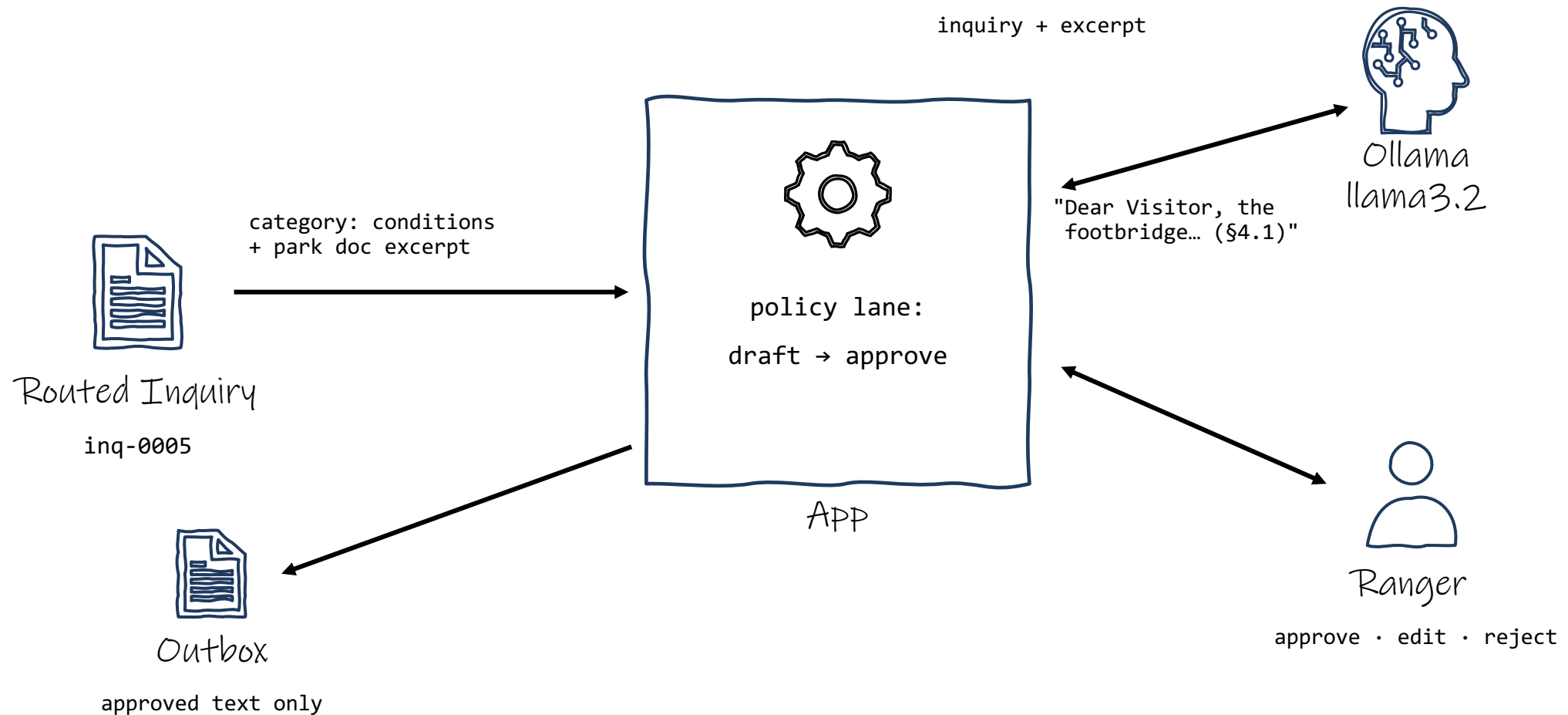
F09 · How It Works



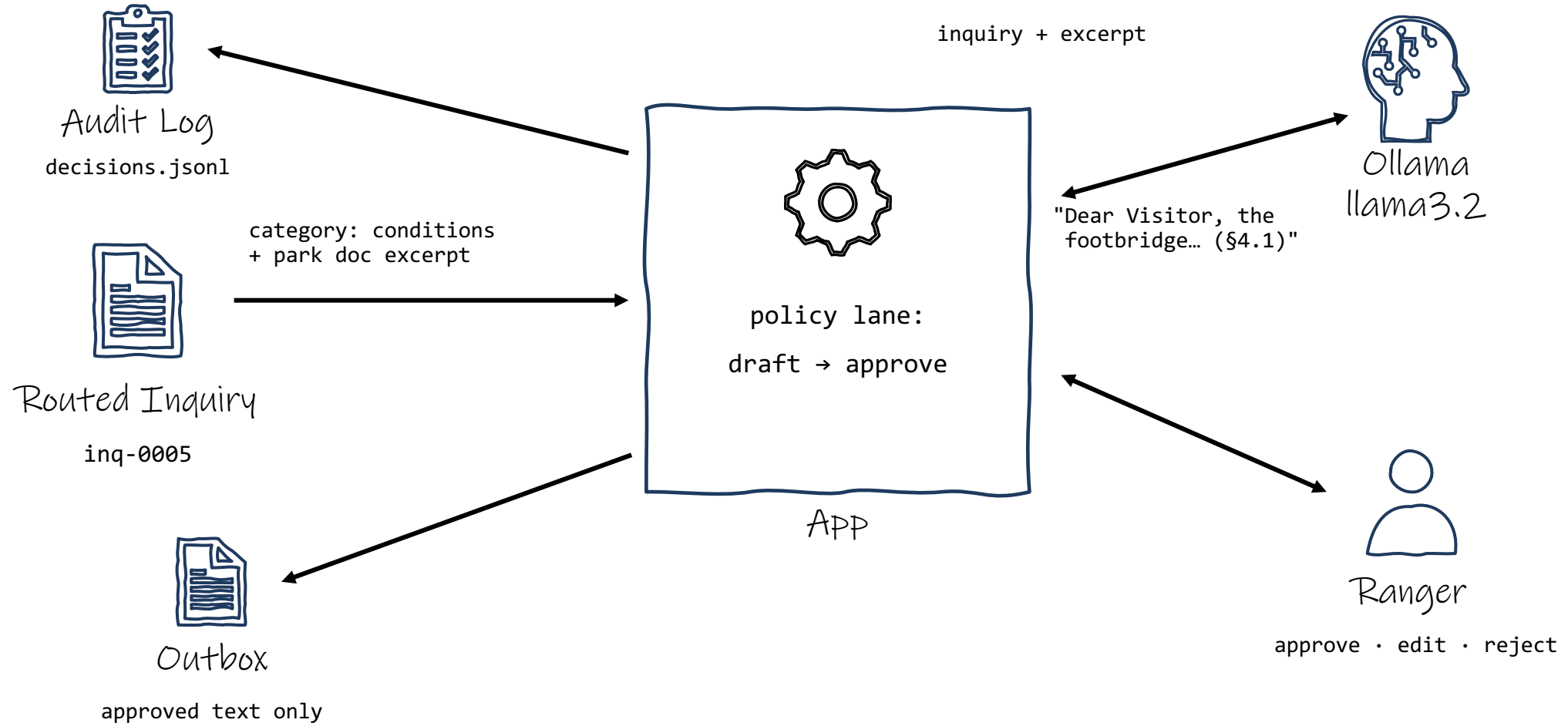
F09 · How It Works



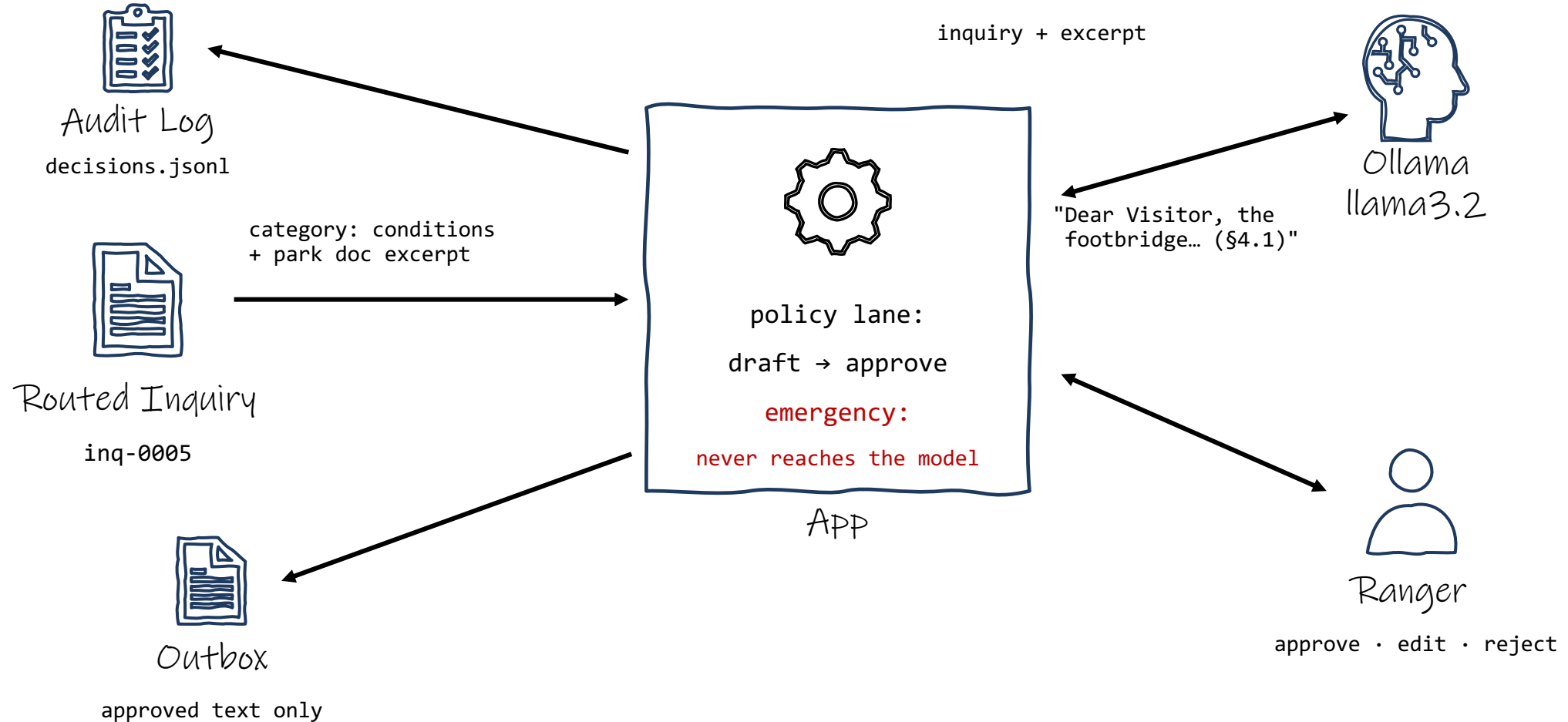
F09 · How It Works



F09 · How It Works



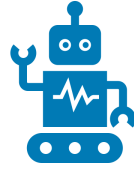
F09 · How It Works



F10 • Agentic Workflows



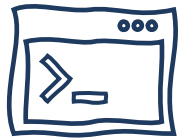
TRAILHEAD
TECHNOLOGY PARTNERS



A model that **picks which tools to call**, in a loop,
until the job is done.

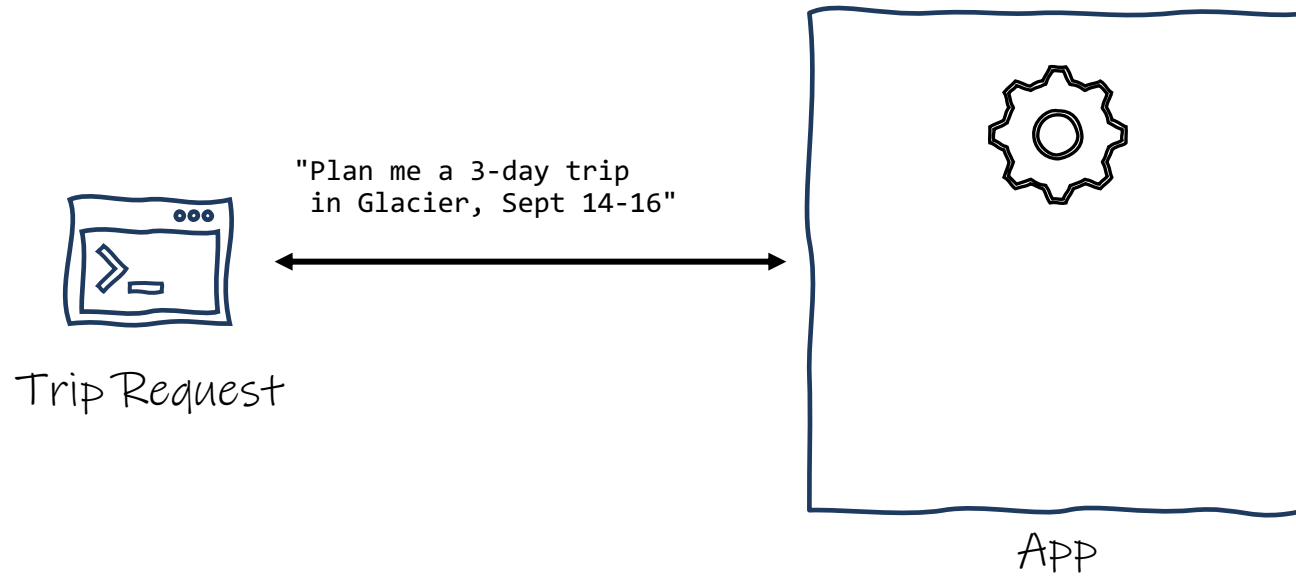


F10 · How the Loop Works

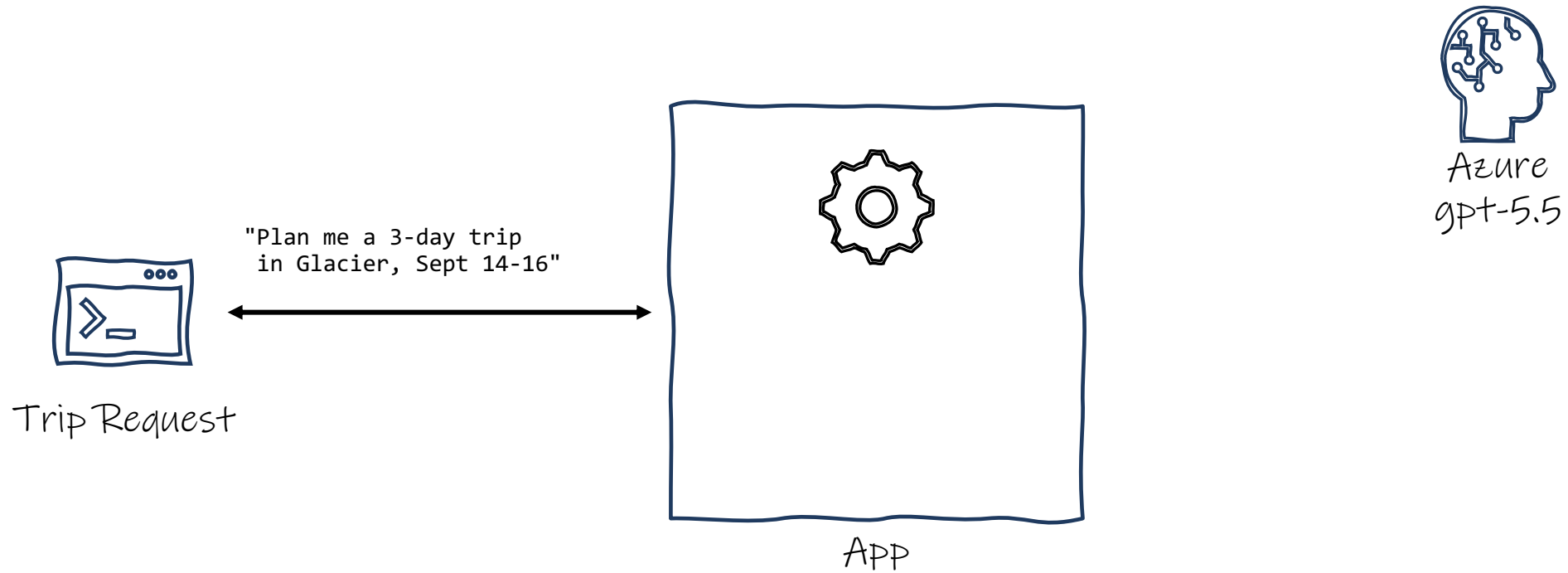


Trip Request

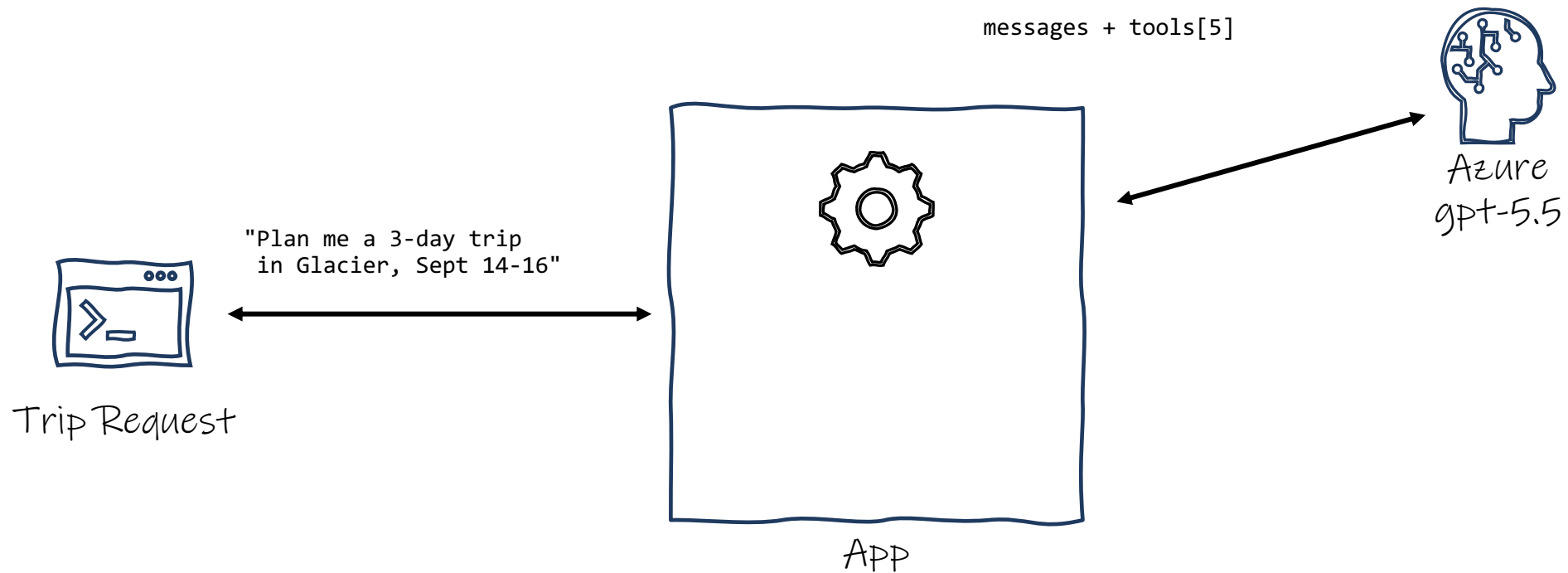
F10 · How the Loop Works



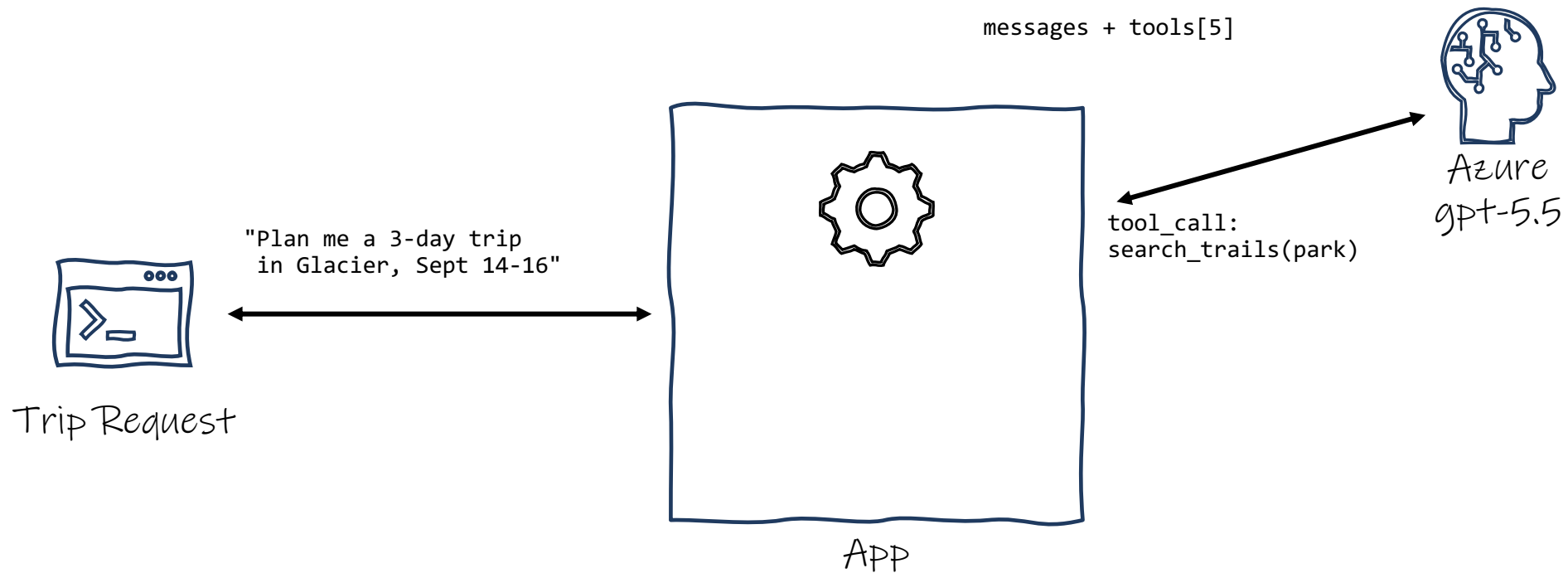
F10 · How the Loop Works



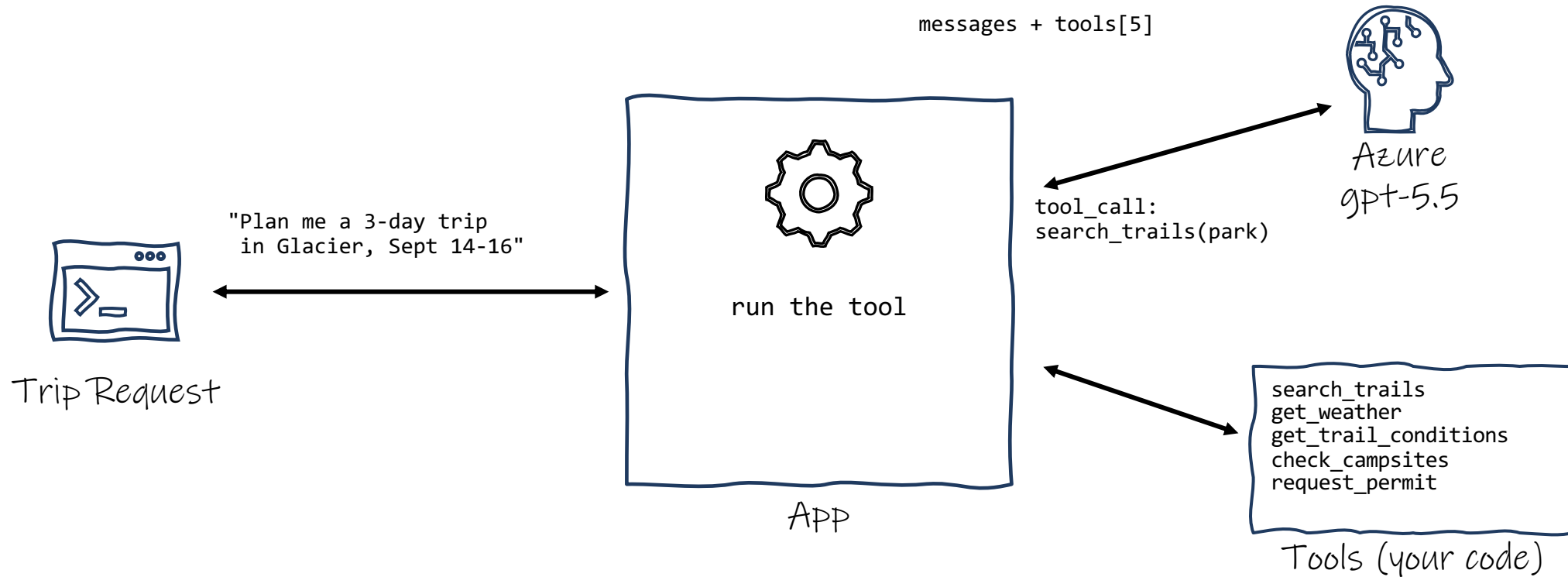
F10 · How the Loop Works



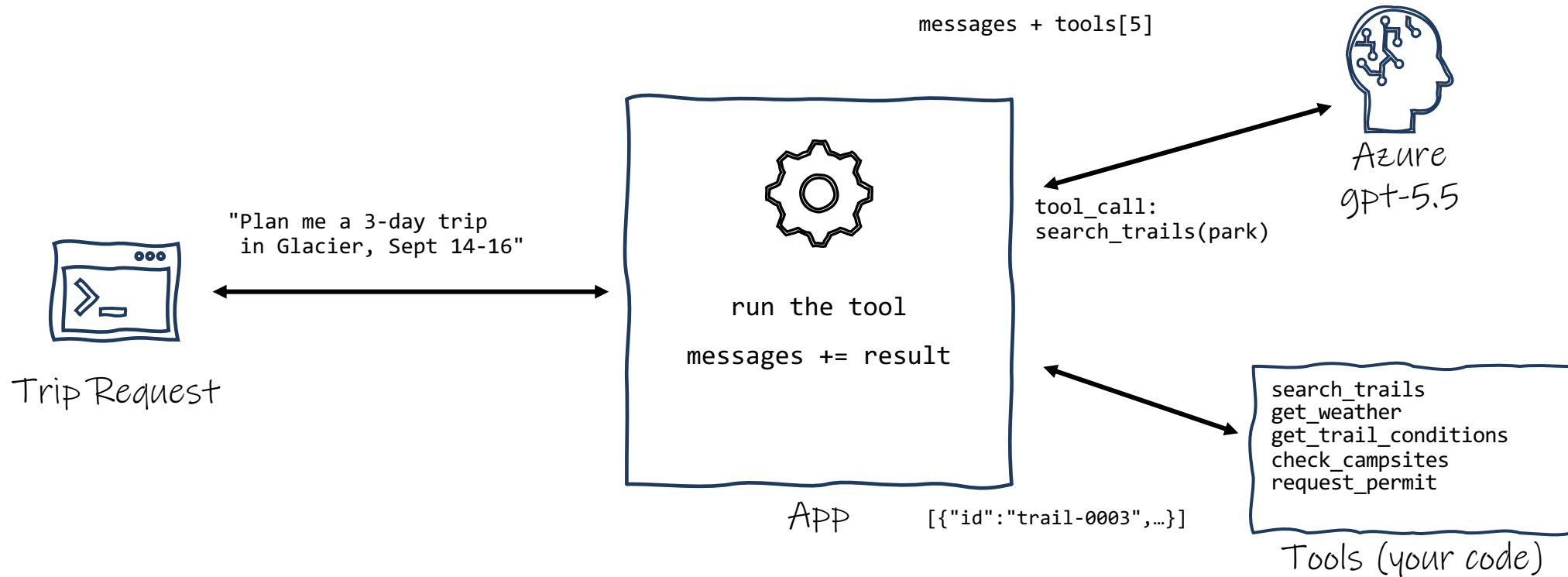
F10 · How the Loop Works



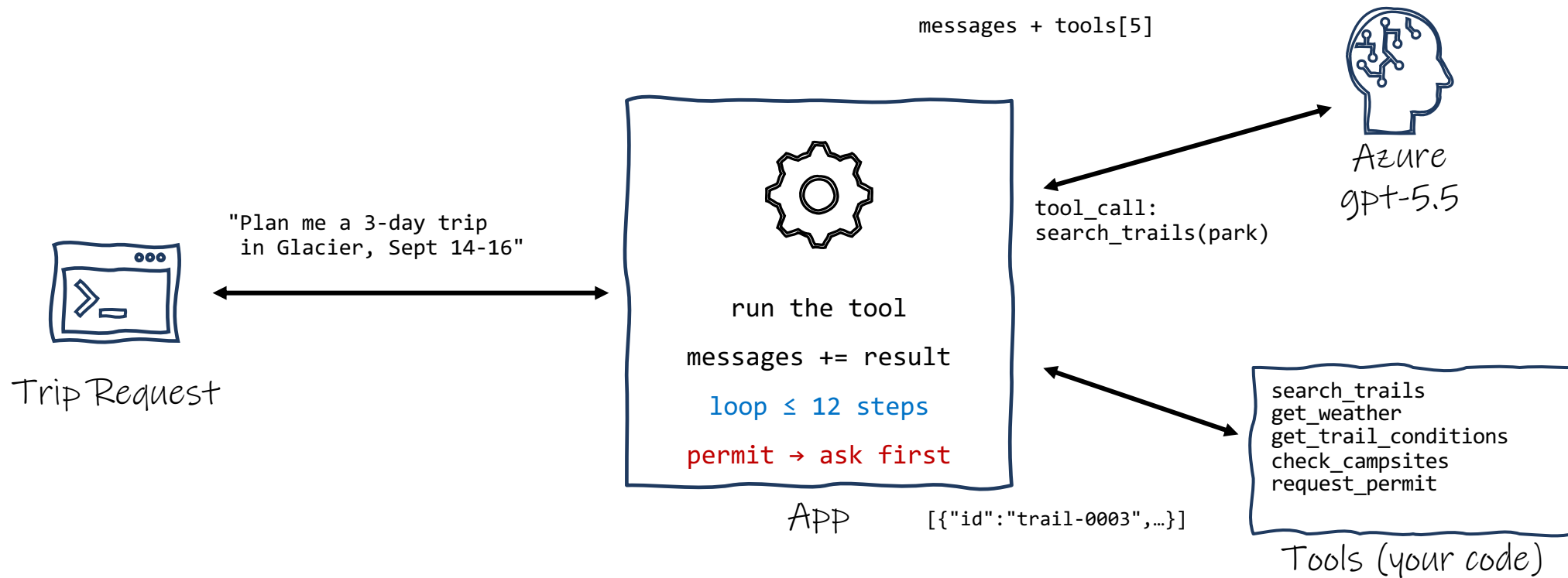
F10 · How the Loop Works



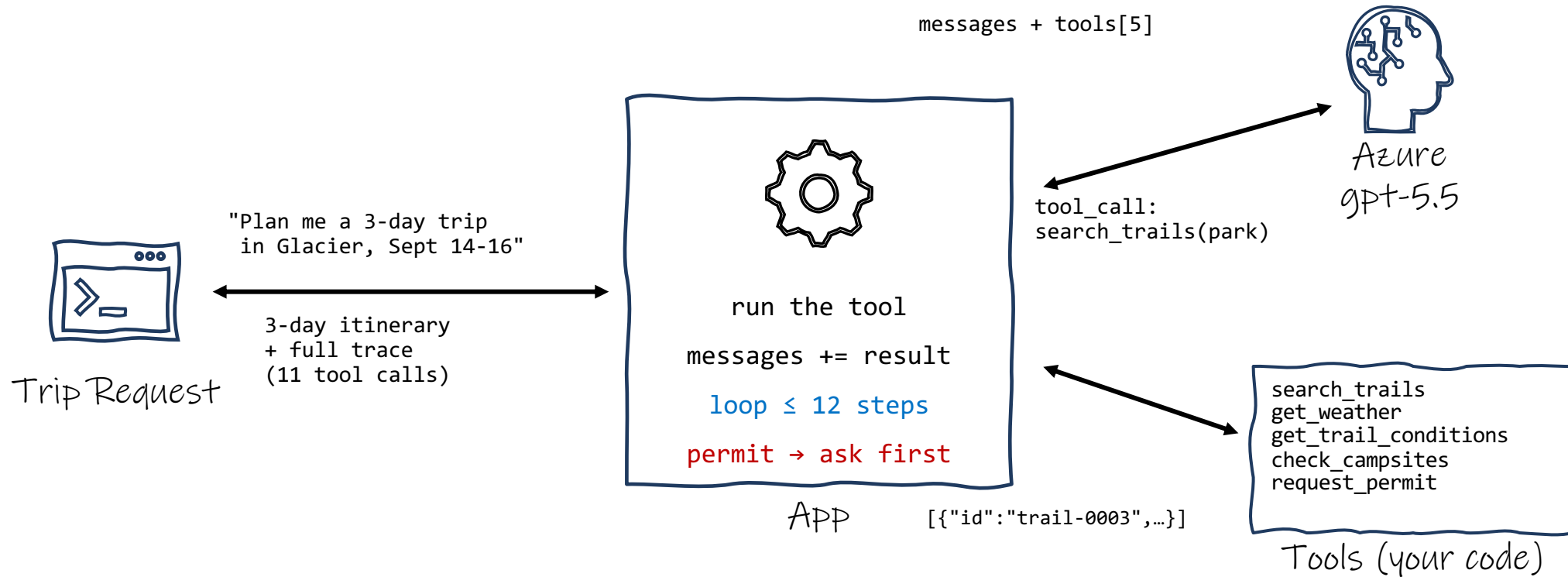
F10 · How the Loop Works



F10 · How the Loop Works



F10 · How the Loop Works



F10 · The Loop Is a Library Call

```
IChatClient client = new ChatClientBuilder(Clients.Cloud("gpt-5.5"))
    // The step budget: the only bound on the loop.
    .UseFunctionInvocation(configure: c => c.MaximumIterationsPerRequest = 12)
    .Build();
var options = new ChatOptions
{
    Tools =
    [
        AIFunctionFactory.Create(SearchTrails),
        AIFunctionFactory.Create(GetWeather),
        AIFunctionFactory.Create(GetTrailConditions),
        AIFunctionFactory.Create(CheckCampsites),
        AIFunctionFactory.Create(RequestPermit),
    ],
};
```

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        AIFunctionFactory.Create(CheckCampsites),
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    ],
};
```

F10 · The Gate

```
[Description("Submit a backcountry permit request. This files a real " +  
    "request, so use it once, at the end, after the plan is settled.")]  
static string RequestPermit(  
    [Description("Park name")] string park,  
    [Description("Permit zone, e.g. 'Lake McDonald / Sperry'")] string zone,  
    [Description("e.g. '2026-09-14 to 2026-09-16'")] string dates,  
    [Description("Number of people")] int groupSize = 2)  
{  
    // The one irreversible action never runs on the model's say-so.  
    if (Gate && !HumanApproves($"File a permit for {zone}, {dates}?"))  
        return Result("""{"status": "cancelled", "message":  
            "The user declined. Do not retry; say no permit was filed."}""");  
  
    return Result(Permits["submit_response"]!.ToJsonString());  
}
```

F10 · The Gate

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```

DEMO · F10 Agentic Workflows

Back to the First Slide

✗ "Where can we use AI?"

Back to the First Slide

✗ "Where can we use AI?"

✓ "What problems can AI best solve for our users?"

Your Move

You've Seen the Patterns, Apply it To Your Systems



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Look For Overlap of Pain and AI Ability

Venn Diagram for the win

Start Small and Easy

Don't try to do too much at once

Start with the easiest and most obvious problem

Safety

Keep a human in the loop until it's trustworthy
Test!

Local vs Cloud Models

Cost vs Accuracy

Keep Trying

<https://github.com/trailheadtechnology/10-ai-features-workshop>

Thanks! Questions?

Jonathan "J." Tower

Founding Partner

🏆 13x Microsoft MVP in .NET & AI

⚙️ VP of .NET Foundation

🎙️ Blue Blazes podcast host

